

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	7
Bill of Material:	9
MDMT Summary:	10
Warnings and Errors:	12
Input Echo:	13
XY Coordinate Calculations:	26
Flg Calc [Int P]: FLANGE	27
Internal Pressure Calculations:	30
Element: Left head	31
Element: Shell-1	33
Element: Shell-2	35
Element: Shell-3	37
Element: Shell-4	39
Element: Shell-5	41
Element: Right head	43
Shop Pressure Test Results:	45
Ring MDMT Calcs:	47
External Pressure Calculations:	48
Element and Detail Weights:	49
Nozzle Flange MAWP:	52
Wind Load Calculation:	54
Earthquake Load Calculation:	56
Center of Gravity Calculation:	58
Saddle Calcs: Operating Case:	60
Saddle Calcs: Test Case:	77
Nozzle Calcs.: N1 40"	91
Nozzle Calcs.: N4 1"	102
Nozzle Calcs.: N5A 1"	105
Nozzle Calcs.: N5B 1"	108
Nozzle Calcs.: N9 8"	111
Nozzle Calcs.: N11 1"	125
Nozzle Calcs.: N2 32"	128
Nozzle Calcs.: N6A 2"	138
Nozzle Calcs.: N6B 2"	141
Nozzle Calcs.: N10 1.5"	144
Nozzle Calcs.: N12A 1"	147
Nozzle Calcs.: N12B 1"	150
Nozzle Calcs.: N3 8"	153
Nozzle Calcs.: N8 3"	167
Nozzle Calcs.: M1 24"	177
Nozzle Calcs.: N7 3"	183



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User (Modify in Config->Save Options)

Job File : ...\\405-V-0037%.pvdb

Date of Analysis : Jul 20, 2026 2:37pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 3500.000 mm OD
 Vessel Design Length, Tangent to Tangent 13000.00 mm
 Specified Datum Line Distance 0.00 mm
 Internal Design Temperature 200 °C
 Internal Design Pressure 1.050 MPa
 External Design Temperature 200 °C
 Maximum Allowable Working Pressure 1 MPa
 Shop Test Pressure 1.425 MPa
 Required Minimum Design Metal Temperature -38.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-516 70	...	...	K02700	
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Rings	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left head	1.057	0.00	1.000	3.0000	No	No
Shell-1	1.057	0.00	1.000	3.0000	N/A	No
Shell-2	1.057	0.00	1.000	3.0000	N/A	No
Shell-3	1.057	0.00	1.000	3.0000	N/A	No
Shell-4	1.057	0.00	1.000	3.0000	N/A	No
Shell-5	1.057	0.00	1.000	3.0000	N/A	No
Right head	1.057	0.00	1.000	3.0000	No	No
N9 blind flg	1.050	0.00	5.000	3.0000	N/A	No

Liquid Level: 1100.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.667

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
2600.00	Bar 250.0 x 20.0	SR1

10400.00 Bar 250.0 x 14.0

SR2

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	20.0	17.0	16.4	...	1.00	1.00
Cylinder	3050.0	3000.0	18.0	18.0	16.5	...	1.00	1.00
Cylinder	6050.0	3000.0	18.0	18.0	16.5	...	1.00	1.00
Cylinder	9050.0	3000.0	18.0	18.0	16.5	...	1.00	1.00
Cylinder	12050.0	3000.0	18.0	18.0	16.5	...	1.00	1.00
Cylinder	12950.0	900.0	18.0	18.0	16.5	...	1.00	1.00
Ellipse	13000.0	50.0	20.0	17.0	16.4	...	1.00	1.00
Body Flg	2800.0	0.0	40.0	39.6	34.1	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1 40"	WRC-297	0.935	Shell	Passed
N9 8"	WRC-297	0.928	Shell	Passed
N2 32"	WRC-297	0.928	Shell	Passed
N3 8"	WRC-297	0.935	Shell	Passed
N8 3"	WRC-297	0.935	Shell	Passed

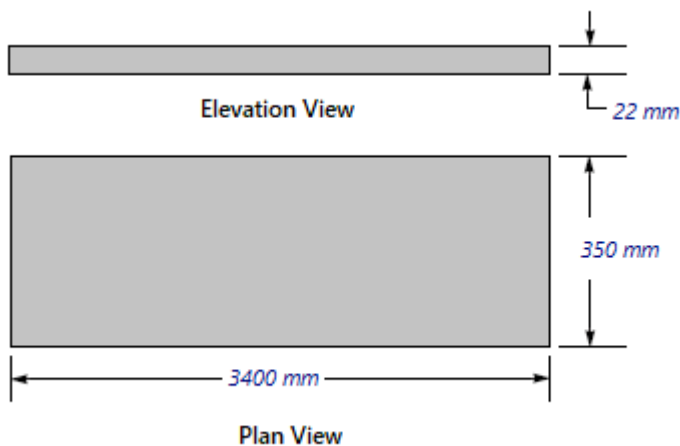
Foundation/Support Summary

Saddle Parameters:

Saddle Width 350.000 mm
 Saddle Bearing Angle 150.000 deg.
 Centerline Dimension 2065.000 mm
 Wear Pad Width 606.000 mm
 Wear Pad Thickness 20.000 mm
 Wear Pad Bearing Angle 162.000 deg.
 Distance from Saddle to Tangent 2600.000 mm

Baseplate Length 3400.000 mm
 Baseplate Thickness 22.000 mm
 Baseplate Width 350.000 mm
 Number of Ribs (including outside ribs) 5
 Rib Thickness 10.000 mm
 Web Thickness 10.000 mm
 Height of Center Web 259.000 mm
 Number of Bolts in Baseplate 4

Baseplate Sketch





Maximum Tensile Bolt Load 3099. N

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	646730.88	N
Maximum Transverse Saddle Shear Load	165027.95	N
Maximum Longitudinal Saddle Shear Load	251950.00	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	920607.50	N
Maximum Transverse Saddle Shear Load	18129.52	N
Maximum Longitudinal Saddle Shear Load	11479.28	N

Weights:

Fabricated - Bare w/o removable internals	34512.4	kgm
Shop Test - Fabricated + water (full)	170813.1	kgm
Shipping - Fab. + rem. intls.+ shipping app.	34512.4	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	44074.4	kgm
Empty - Fab. + intls. + details + wghts.	44074.4	kgm
Operating - Empty + operating liquid (no ca)	68255.0	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1 40"	1	...	...	OK	11.33	OK	Passed
N4 1"	1	...	...	OK	6.42	OK	No Calc[*]
N5A 1"	1	...	...	OK	6.42	OK	No Calc[*]
N5B 1"	1	...	...	OK	6.42	OK	No Calc[*]
N9 8"	1	...	...	OK	10.16	OK	Passed
N11 1"	1	...	...	OK	6.42	OK	No Calc[*]
N2 32"	1	...	...	OK	11.33	OK	Passed
N6A 2"	1	...	...	OK	7.80	OK	No Calc[*]
N6B 2"	1	...	...	OK	7.80	OK	No Calc[*]
N10 1.5"	1	...	...	OK	7.52	OK	No Calc[*]
N12A 1"	1	...	...	OK	6.42	OK	No Calc[*]
N12B 1"	1	...	...	OK	6.42	OK	No Calc[*]
N3 8"	1	...	...	OK	10.16	OK	Passed
N8 3"	1	...	...	OK	8.73	OK	Passed
M1 24"	1	...	...	...		OK	Passed
N7 3"	4	...	...	OK	8.26	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1 Nozzle : M1 24"
 Minimum MAWP Shells/Flanges : 1 Element : Right head
 Minimum MAPnc Shells/Flanges : 1 Element : Right head

Computed Vessel M.A.W.P. : 1 MPa

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N1 40"	1160.000	90.000	2044.000
20	N4 1"	2300.000	90.000	80.800
20	N5A 1"	1000.000	-16.428	80.800
20	N5B 1"	1000.000	270.000	80.800
20	N9 8"	2800.000	90.000	399.350
20	N11 1"	2450.000	90.000	80.800
50	N2 32"	12039.998	90.000	1637.600
50	N6A 2"	11700.000	-16.428	113.600
50	N6B 2"	11700.000	270.000	113.600
50	N10 1.5"	11999.999	-59.605	96.548
50	N12A 1"	11399.999	-16.428	80.800
50	N12B 1"	11399.999	270.000	80.800
60	N3 8"	12359.999	270.000	399.350
60	N8 3"	12769.999	270.000	164.400
70	M1 24"	0.000	0.000	1231.200
80	N7 3"	0.000	0.000	180.944

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

Checking Multiple Nozzles on Flat Head per ASME Sec. VIII Div. 1 UG-39

Comparing Nozzles on Element: N9 blind flg

Note: No Nozzle pairs found on this element.

UG-39 Nozzle Diameter and Distance to Edge Checks:

Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
N7 3"	82.2	131.83	136.65	20.55

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N4 1"	1.000 in	Actual	LWN	2.000	12.700	...	...	318.18	150
N5A 1"	1.000 in	Actual	LWN	2.000	12.700	...	...	326.56	150
N5B 1"	1.000 in	Actual	LWN	2.000	12.700	...	...	268.18	150
N11 1"	1.000 in	Actual	LWN	2.000	12.700	...	...	318.18	150
N12A 1"	1.000 in	Actual	LWN	2.000	12.700	...	...	276.56	150
N12B 1"	1.000 in	Actual	LWN	2.000	12.700	...	...	268.18	150
N10 1.5"	1.500 in	Actual	LWN	2.620	14.224	...	...	348.12	150
N6A 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	280.85	150
N6B 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	268.43	150
N8 3"	3.000 in	Actual	LWN	4.620	20.574	...	...	268.98	300
N7 3"	3.000 in	Actual	LWN	4.240	15.748	...	...	289.67	150
N9 8"	8.000 in	80	WNF	8.625	12.700	359.08	18.00	271.43	300
N3 8"	8.000 in	80	WNF	8.625	12.700	359.08	18.00	271.43	300
M1 24"	24.000 in	Actual	WNF	24.945	12.000	1033.60	18.00	282.89	150
N2 32"	32.000 in	Actual	WNF	33.575	20.000	...	...	584.74	300
N1 40"	40.000 in	Actual	WNF	42.520	32.000	...	...	421.40	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:
 Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N4 1"	SA-350 LF2	18.000	10.000	...	...	...
N5A 1"	SA-350 LF2	18.000	10.000	...	...	...
N5B 1"	SA-350 LF2	18.000	10.000	...	...	...
N11 1"	SA-350 LF2	18.000	10.000	...	...	...
N12A 1"	SA-350 LF2	18.000	10.000	...	...	...
N12B 1"	SA-350 LF2	18.000	10.000	...	...	...
N10 1.5"	SA-350 LF2	18.000	10.000	...	...	...
N6A 2"	SA-350 LF2	18.000	10.000	...	...	...
N6B 2"	SA-350 LF2	18.000	10.000	...	...	...
N8 3"	SA-350 LF2	18.000	10.000	...	...	...
N7 3"	SA-350 LF2	39.624	10.000	...	...	...
N9 8"	SA-333 6	18.000	10.000	14.000	18.000	...
N3 8"	SA-333 6	18.000	10.000	14.000	18.000	...
M1 24"	SA-516 70	18.000	10.000	14.000	18.000	...
N2 32"	SA-516 70	18.000	10.000	...	...	...
N1 40"	SA-516 70	24.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N4 1"	2300.000	90.0	300.00	0.00	Shell-1
N5A 1"	1000.000	-16.4	300.00	0.00	Shell-1
N5B 1"	1000.000	270.0	250.00	0.00	Shell-1
N11 1"	2450.000	90.0	300.00	0.00	Shell-1
N12A 1"	11399.999	-16.4	250.00	0.00	Shell-4



N12B 1"	11399.999	270.0	250.00	0.00	Shell-4
N10 1.5"	11999.998	-59.6	250.00	0.00	Shell-4
N6A 2"	11699.999	-16.4	250.00	0.00	Shell-4
N6B 2"	11699.999	270.0	250.00	0.00	Shell-4
N8 3"	12769.999	270.0	250.00	0.00	Shell-5
N7 3"	...	0.0	250.00	0.00	N9 blind flg
N9 8"	2800.000	90.0	250.00	0.00	Shell-1
N3 8"	12359.999	270.0	250.00	0.00	Shell-5
M1 24"	...	0.0	250.00	0.00	Right head
N2 32"	12039.998	90.0	500.00	0.00	Shell-4
N1 40"	1160.000	90.0	300.00	0.00	Shell-1

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 20.0mm THK X 3500.0mm ID X 50.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 20.0mm THK X 3502.0mm ID X 50.0mm	SA-516 70
1	BLIND FLANGE: 40.0mm THK X 381.0mm OD	SA-516 70
1	PLATFORM: 1000mm X 3536mm WIDE	
1	LADDER: 4000.0mm	...
2	INSULATION: 925mm X 45mm THK	
2	SADDLE: 350.0mm X 150 DEG	SA-516 70
4	INSULATION: 3000mm X 45mm THK	
1	BAR RING STIFFENER: 3536.0mm ID X 4036.0mm OD X 20.0mm	SA-516 70
1	BAR RING STIFFENER: 3536.0mm ID X 4036.0mm OD X 14.0mm	SA-516 70
1	INSULATION: 900mm X 45mm THK	
1	INSULATION: 39mm X 45mm THK	
1	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
1	GASKET: 263.7mm OD X 233.4mm ID	...
12	BODY FLANGE BOLTS: 22.2mm DIA	SA-193 B7
12	NUTS FOR BODY FLANGE BOLTS: 22.2mm DIA	...
2	CYLINDRICAL SEGMENT 18.0mm THK X 4374.9mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 2245.5mm X 3000.0mm	SA-516 70
9	CYLINDRICAL SEGMENT 18.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 1851.5mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 18.0mm THK X 7947.4mm X 3000.0mm	SA-516 70
3	CYLINDRICAL SEGMENT 18.0mm THK X 3048.0mm X 900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 7947.4mm X 900.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
N9 blind flg	[11]	B	-29	-104	-29	0.172	9.906	1.00	No
Left head	[10]	B	-12	-12	-29	0.999	17.000	1.00	No
Left head	[7]	B	-7	-17	-29	0.828	20.000	1.00	No
Shell-1	[8]	B	-11	-14	-29	0.939	18.000	1.00	No
Shell-2	[8]	B	-11	-14	-29	0.939	18.000	1.00	No
Shell-3	[8]	B	-11	-14	-29	0.939	18.000	1.00	No
Shell-4	[8]	B	-11	-14	-29	0.939	18.000	1.00	No
Shell-5	[8]	B	-11	-14	-29	0.939	18.000	1.00	No
Right head	[10]	B	-12	-12	-29	1.000	17.000	1.00	No
Right head	[7]	B	-7	-17	-29	0.829	20.000	1.00	No
SR1	[23]	B	-11	-14	-29	0.939	18.000	1.00	No
SR2	[23]	B	-18	-22	-29	0.939	14.000	1.00	No
Left Saddle	[24]	B	-11	-14	-29	0.939	18.000	1.00	No
22	[24]	B	-11	-14	-29	0.939	18.000	1.00	No
N1 40"	[1]	B	-38	-104		0.141	32.000	1.00	No
Nozzle Flg	[5]	!	-46	-104					
N4 1"	[1]	B	-21	-104	-29	0.013	12.700	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N5A 1"	[1]	B	-21	-104	-29	0.013	12.700	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N5B 1"	[1]	B	-21	-104	-29	0.013	12.700	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N9 8"	[1]	B	-11	-29	-29	0.933	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-104					
N11 1"	[1]	B	-21	-104	-29	0.013	12.700	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N2 32"	[1]	B	-38	-104		0.192	20.000	1.00	No
Nozzle Flg	[5]	!	-46	-104					
N6A 2"	[1]	B	-19	-104	-29	0.022	13.462	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N6B 2"	[1]	B	-19	-104	-29	0.022	13.462	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N10 1.5"	[1]	B	-18	-104	-29	0.016	14.224	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N12A 1"	[1]	B	-21	-104	-29	0.013	12.700	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N12B 1"	[1]	B	-21	-104	-29	0.013	12.700	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N3 8"	[1]	B	-11	-29	-29	0.939	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-104					
N8 3"	[1]	B	-11	-29	-29	0.939	18.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
M1 24"	[1]	B	-12	-29	-29	0.895	17.000	1.00	No
Nozzle Flg	[5]	!	-46	-71					
N7 3"	[1]	B	-15	-104	-29	0.026	15.748	1.00	No
Nozzle Flg	[5]	!	-46	-46					
Bolting	[21]		-48						

Warmest MDMT: -7 -12

Required Minimum Design Metal Temperature -38.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.



- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	1.05	MPa
Design Internal Temperature	200.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-38.0	°C
Type of Construction	Welded	
Special Service	Amine	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.081
0.050	0.123
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.104
0.800	0.088
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000



Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	1.500
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.081
0.050	0.123
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.104
0.800	0.088
1.000	0.070
1.200	0.058
1.400	0.050



1.600 0.044
 1.800 0.039
 2.000 0.035
 2.400 0.029
 0.000 0.000

Consider Vertical Acceleration Yes
 M.A.W.P + Static Head to Nozzle Y
 Consider MAP New and Cold in Noz. Design N
 Consider External Loads for Nozzle Des. Y
 Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element# 1/8
 Element From Node 10
 Element To Node 20
 Element Type Elliptical
 Description Left head
 Distance "FROM" to "TO" 50 mm
 Inside Diameter 3500 mm
 Element Thickness 17 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 20 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 1.05 MPa
 Design Temperature Internal Pressure 200 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 200 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Allowable Stress, Ambient 138 MPa
 Allowable Stress, Operating 138 MPa
 Allowable Stress, Hydrotest 234 MPa
 Material Density 0.00775 kg/cm³
 P Number Thickness 32.004 mm
 Yield Stress, Operating 225 MPa
 UCS-66 Chart Curve Designation B
 External Pressure Chart Name CS-2
 UNS Number K02700
 Product Form Plate
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No
 Element From Node 10
 Detail Type Platform
 Detail ID PLATFORM



Dist. from "FROM" Node / Offset dist	50	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	1704000	mm ²
Platform Weight	78149	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	3536	mm
Platform Height	1000	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	13000	mm

Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1100	mm
Liquid Density	0.0006662	kg/cm ³

Element From Node	10	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	-875	mm
Height/Length of Insulation	925	mm
Thickness of Insulation	45	mm
Density	0.00012	kg/cm ³

Element#	2/8	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	3500	mm
Element Thickness	18	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.05	MPa
Design Temperature Internal Pressure	200	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	2550	mm
Width of Saddle	350	mm
Height of Saddle at Bottom	2065	mm
Saddle Contact Angle	150.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	606	mm
Thickness of Wear Plate	20	mm
Contact Angle, Wear Plate (degrees)	162.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	



Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1100	mm
Liquid Density	0.0006662	kg/cm ³
Element From Node	20	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	45	mm
Density	0.00012	kg/cm ³
Element From Node	20	
Detail Type	Ring	
Detail ID	SR1	
Dist. from "FROM" Node / Offset dist	2550	mm
Inside Diameter of Ring	3536	mm
Thickness of Ring	20	mm
Outside Diameter of Ring	4036	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N1 40"	
Dist. from "FROM" Node / Offset dist	1110	mm
Nozzle Diameter	40	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	8583.9	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N4 1"	
Dist. from "FROM" Node / Offset dist	2250	mm
Nozzle Diameter	1	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	73.33	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N5A 1"	
Dist. from "FROM" Node / Offset dist	950	mm
Nozzle Diameter	1	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	-16.4277	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	73.33	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	20	
Detail Type	Nozzle	



Detail ID	N5B 1"
Dist. from "FROM" Node / Offset dist	950 mm
Nozzle Diameter	1 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	44.445 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	N9 8"
Dist. from "FROM" Node / Offset dist	2750 mm
Nozzle Diameter	8 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	923.38 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	N11 1"
Dist. from "FROM" Node / Offset dist	2400 mm
Nozzle Diameter	1 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	73.33 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	20
Detail Type	Weight
Detail ID	N1 Insert plate
Dist. from "FROM" Node / Offset dist	1150 mm
Miscellaneous Weight	4903 N
Offset from Element Centerline	1750 mm

Element From Node	20
Detail Type	Weight
Detail ID	Fy wt-1
Dist. from "FROM" Node / Offset dist	2550 mm
Miscellaneous Weight	245.15 N
Offset from Element Centerline	0 mm

Element From Node	20
Detail Type	Weight
Detail ID	Lifting lug 1
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	2255.4 N
Offset from Element Centerline	1750 mm

Element From Node	20
Detail Type	Weight
Detail ID	Lifting lug 2
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	2255.4 N
Offset from Element Centerline	1750 mm

Element From Node	20
Detail Type	For./Mom.
Detail ID	Nozzles F/M 1
Dist. from "FROM" Node / Offset dist	2550 mm
Force in X Direction	125975 N



Force in Y Direction	0	N
Force in Z Direction	110090	N
Moment about X Axis	0	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	0	N-mm
Force/Moment Combination Method	SRSS	

Element#	3/8
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3500 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.05 MPa
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1100 mm
Liquid Density	0.0006662 kg/cm³

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	45 mm
Density	0.00012 kg/cm³

Element From Node	30
Detail Type	Weight
Detail ID	Cleat weight
Dist. from "FROM" Node / Offset dist	510 mm
Miscellaneous Weight	11277 N
Offset from Element Centerline	0 mm

Element#	4/8
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3500 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.05 MPa
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2



Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 40
 Detail Type Liquid
 Detail ID Liquid 40
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 1100 mm
 Liquid Density 0.0006662 kg/cm³

Element From Node 40
 Detail Type Insulation
 Detail ID Ins: 40
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 3000 mm
 Thickness of Insulation 45 mm
 Density 0.00012 kg/cm³

Element# 5/8
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description Shell-4
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 3500 mm
 Element Thickness 18 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 18 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 1.05 MPa
 Design Temperature Internal Pressure 200 °C
 Design External Pressure 0 MPa
 Design Temperature External Pressure 200 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Saddle
 Detail ID 22
 Dist. from "FROM" Node / Offset dist 1350 mm
 Width of Saddle 350 mm
 Height of Saddle at Bottom 2065 mm
 Saddle Contact Angle 150.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 606 mm
 Thickness of Wear Plate 20 mm
 Contact Angle, Wear Plate (degrees) 162.0
 Friction coefficient 0.1
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 mm

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid 50
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 1100 mm
 Liquid Density 0.0006662 kg/cm³

Element From Node 50
 Detail Type Insulation



Detail ID Ins: 50
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 3000 mm
 Thickness of Insulation 45 mm
 Density 0.00012 kg/cm³

Element From Node 50
 Detail Type Ring
 Detail ID SR2
 Dist. from "FROM" Node / Offset dist 1350 mm
 Inside Diameter of Ring 3536 mm
 Thickness of Ring 14 mm
 Outside Diameter of Ring 4036 mm
 Material Name SA-516 70
 Height of Section Ring 0 mm
 Using Custom Stiffener Section No

Element From Node 50
 Detail Type Nozzle
 Detail ID N2 32"
 Dist. from "FROM" Node / Offset dist 2990 mm
 Nozzle Diameter 32 in.
 Nozzle Schedule None
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 5059.1 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-516 70 +IT

Element From Node 50
 Detail Type Nozzle
 Detail ID N6A 2"
 Dist. from "FROM" Node / Offset dist 2650 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle -16.4277
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 78.097 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 50
 Detail Type Nozzle
 Detail ID N6B 2"
 Dist. from "FROM" Node / Offset dist 2650 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 78.097 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 50
 Detail Type Nozzle
 Detail ID N10 1.5"
 Dist. from "FROM" Node / Offset dist 2950 mm
 Nozzle Diameter 1.5 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle -59.6049
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 69.007 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 50



Detail Type	Nozzle	
Detail ID	N12A 1"	
Dist. from "FROM" Node / Offset dist	2350	mm
Nozzle Diameter	1	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	-16.4277	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	44.447	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	50	
Detail Type	Nozzle	
Detail ID	N12B 1"	
Dist. from "FROM" Node / Offset dist	2350	mm
Nozzle Diameter	1	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	44.447	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	50	
Detail Type	Weight	
Detail ID	N2 insert pleat	
Dist. from "FROM" Node / Offset dist	2990	mm
Miscellaneous Weight	2275	N
Offset from Element Centerline	1750	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Fy wt-2	
Dist. from "FROM" Node / Offset dist	1350	mm
Miscellaneous Weight	245.15	N
Offset from Element Centerline	0	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Lifting lug 3	
Dist. from "FROM" Node / Offset dist	2900	mm
Miscellaneous Weight	2255.4	N
Offset from Element Centerline	1750	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Lifting lug 4	
Dist. from "FROM" Node / Offset dist	2900	mm
Miscellaneous Weight	2255.4	N
Offset from Element Centerline	1750	mm
Element From Node	50	
Detail Type	For./Mom.	
Detail ID	Nozzle F/M 2	
Dist. from "FROM" Node / Offset dist	1350	mm
Force in X Direction	125975	N
Force in Y Direction	0	N
Force in Z Direction	110090	N
Moment about X Axis	0	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	0	N-mm
Force/Moment Combination Method	SRSS	



Element Type	Cylinder
Description	Shell-5
Distance "FROM" to "TO"	900 mm
Inside Diameter	3500 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.05 MPa
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1100 mm
Liquid Density	0.0006662 kg/cm³
Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	900 mm
Thickness of Insulation	45 mm
Density	0.00012 kg/cm³
Element From Node	60
Detail Type	Nozzle
Detail ID	N3 8"
Dist. from "FROM" Node / Offset dist	310 mm
Nozzle Diameter	8 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	580.06 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT
Element From Node	60
Detail Type	Nozzle
Detail ID	N8 3"
Dist. from "FROM" Node / Offset dist	720 mm
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	183.62 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT
Element From Node	60
Detail Type	Weight
Detail ID	N3 Vortex break
Dist. from "FROM" Node / Offset dist	310 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	1600 mm



Element To Node	80	
Element Type	Elliptical	
Description	Right head	
Distance "FROM" to "TO"	50	mm
Element Outside Diameter	3536	mm
Element Thickness	17	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.05	MPa
Design Temperature Internal Pressure	200	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1100	mm
Liquid Density	0.0006662	kg/cm³
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	925	mm
Thickness of Insulation	45	mm
Density	0.00012	kg/cm³
Element From Node	70	
Detail Type	Nozzle	
Detail ID	M1 24"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	4164.2	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	
Element From Node	70	
Detail Type	Weight	
Detail ID	M1 davit	
Dist. from "FROM" Node / Offset dist	800	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	0	mm

Element#	8/8	
Element From Node	80	
Element To Node	90	
Element Type	Flange	
Description	N9 blind flg	
Distance "FROM" to "TO"	39.624	mm
Flange Outside Diameter	381	mm
Element Thickness	39.624	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	40	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.05	MPa
Design Temperature Internal Pressure	200	°C



Design External Pressure	0	MPa
Design Temperature External Pressure	200	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	39.624	mm
Thickness of Insulation	45	mm
Density	0.00012	kg/cm ³
Element From Node	80	
Detail Type	Nozzle	
Detail ID	N7 3"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	139.36	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Left head		50	...	50	...
Shell-1		3050	...	3000	...
Shell-2		6050	...	3000	...
Shell-3		9050	...	3000	...
Shell-4		12050	...	3000	...
Shell-5		12950	...	900	...
Right head		13000	...	50	...
N9 blind flg		2800	2000	...	39.624

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

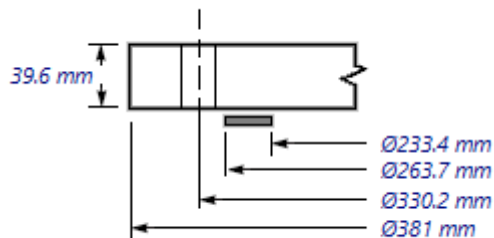
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Flange Input Data Values Description: FLANGE :

N9 blind flg

Flange Type		Blind	
Design Pressure	P	1.05	MPa
Design Temperature		200	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	381.000	mm
Flange Thickness	t	39.6240	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Perimeter along the Center of the Bolts	L	1037.354	mm
Diameter of Bolt Circle	C	330.200	mm
Nominal Bolt Diameter	a	22.2250	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		12	
Flange Face Outside Diameter	Fod	269.748	mm
Flange Face Inside Diameter	Fid	184.150	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	263.652	mm
Gasket Inside Diameter	Gi	233.426	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		-2000.00	N
Bending Moment on Flange		1081200.00	N-mm

External forces and moments are not considered in determining required bolt areas for bolted covers in Part 4.16 flange analysis, see Table 4.16.7.



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = Self-Energizing	0.000	mm
Basic Gasket Width,	bo = N/2	0.000	mm
Effective Gasket Width,	b = bo	0.000	mm
Gasket Reaction Diameter,	G = Go (Self-Energizing)	263.652	mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 263.652^2 \cdot 1.05$$

$$= 57319.766 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 0.0 \cdot \pi \cdot 263.652 \cdot 0.0 \cdot 1.05$$

$$= 0.000 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(57320 + 0 + 0, 0)$$

$$= 57319.766 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 0.0 \cdot 0.0 \cdot \pi \cdot 263.652 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 0.000 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$$

$$= \max((57320 + -2000 + 4 \cdot 1081200 / 263.65) / 172, 0 / 172)$$

$$= 428.629 \text{ mm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	428.629	3243.864	
Radial Distance between Bolts and Edge:	23.812	25.400	
Circ. Spacing between the Bolts:	52.400	85.462	...

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 3243.8643$$

$$= 557887.19 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 57319.77 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (330.2 - 263.652) / 2$$

$$= 33.2740 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 57320 \cdot 33.274 \cdot 1.0 / (36.624^2 \cdot 263.652) + 0.3 \cdot 1.0 \cdot 1.05 \cdot 263.652^2 / 36.624^2$$

$$= 26.57 \text{ MPa}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 557887 \cdot 33.274 \cdot 1.0 / (36.624^2 \cdot 263.652)$$

$$= 99.74 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= (Wm1 + Fa + 4Me/G) / Ab$$

$$= 75717 / 3243.8643$$

$$= 23.34 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$



= (0/3243.8643)
 = 0.00 MPa

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	26.57	138.00	99.74 138.00
Bolting	23.34	172.00	0.00 172.00

Reqd. Blind Flange Thickness at Center 34.137 mm
 Reqd. Blind Flange Thickness at Gasket 34.137 mm

Estimated M.A.W.P. (Operating) 5 MPa
 Estimated Finished Weight of Flange at given Thk. 35.0 kgm
 Estimated Unfinished Weight of Forging at given Thk 35.0 kgm

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.172, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	22.70	138.00	85.21 138.00
Bolting	23.34	172.00	0.00 172.00

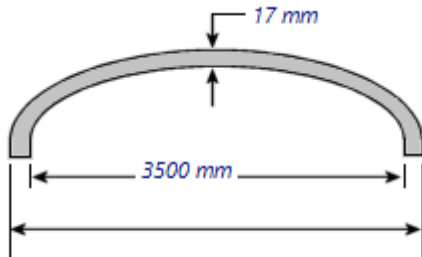
Minimum Req'd Thickness, ambient, no Corrosion: 31.140 mm

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Left head

Elliptical Head Design Information:					
Design Pressure	P	1.057 MPa	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3500.000 mm	Actual Stress		132.2 MPa
Spec. Min. Thk t,e, or sa		17.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		14140132.0 mm²	SG of Contents		0.667
Element Volume		6094.5 ltr	Weight of Contents		9335.2 N
Empty Weight		23381.7 N	Operating Weight		32716.9 N
			Comp. All. Ope. Stress		92.9 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.875 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.057 \cdot 3506.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 1.057)$$

$$= 13.4096 + 3.0000 = 16.4096 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 14.0) / (0.998 \cdot 3506.0 + 0.2 \cdot 14.0)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 17.0) / (1.0 \cdot 3500.0 + 0.2 \cdot 17.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1.057 \cdot (0.998 \cdot 3506.0 + 0.2 \cdot 14.0)) / (2 \cdot 1.0 \cdot 14.0)$$

$$= 132.183 \text{ MPa}$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (1.057 \cdot 1753.0) / (138.0 \cdot 1.0 - 0.6 \cdot 1.057) + 3.0$
 $= 16.492 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.007 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 1.0 \cdot 17.0) / (1753.0 + 0.6 \cdot 17.0)$
 $= 1 - 0.01 = 1 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (3506.0 / (2 \cdot 878.0))^2) / 6$
 $= 0.997724$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.479 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 17.0, tr = 13.992, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.999$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -12 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 20.0, tr = 14.08, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.828$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -17 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.103 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.828	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

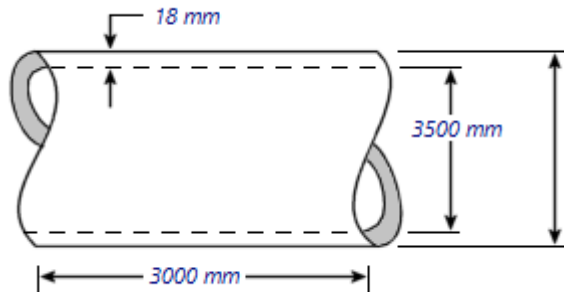
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.875 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	1.057 MPa	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3500.000 mm	Actual Stress		124.2 MPa
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		33326016.0 mm ²	SG of Contents		0.667
Element Volume		28868.6 ltr	Weight of Contents		50799.6 N
Empty Weight		45359.1 N	Operating Weight		96158.6 N
			Comp. All. Ope. Stress		94.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.057 \cdot 1753.0) / (138.0 \cdot 1.0 - 0.6 \cdot 1.057)$$

$$= 13.4920 + 3.0000 = 16.4920 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (1753.0 + 0.6 \cdot 15.0)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (1750.0 + 0.6 \cdot 18.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.057 \cdot (1753.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 124.187 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.512 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0



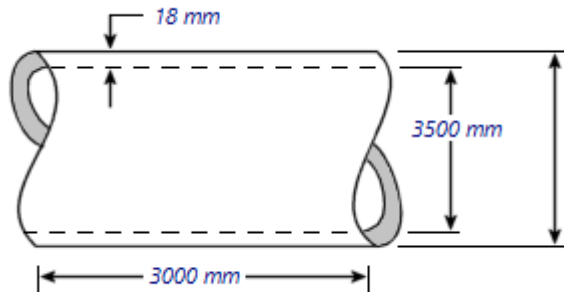
Thickness Ratio = $tr * E^* / (tg - c) = 0.939$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.103 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.939	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	-14 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	1.057 MPa	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3500.000 mm	Actual Stress		124.2 MPa
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		33326016.0 mm²	SG of Contents		0.667
Element Volume		28868.6 ltr	Weight of Contents		50799.6 N
Empty Weight		45359.1 N	Operating Weight		96158.6 N
			Comp. All. Ope. Stress		94.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.057 \cdot 1753.0) / (138.0 \cdot 1.0 - 0.6 \cdot 1.057)$$

$$= 13.4920 + 3.0000 = 16.4920 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (1753.0 + 0.6 \cdot 15.0)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (1750.0 + 0.6 \cdot 18.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.057 \cdot (1753.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 124.187 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.512 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0



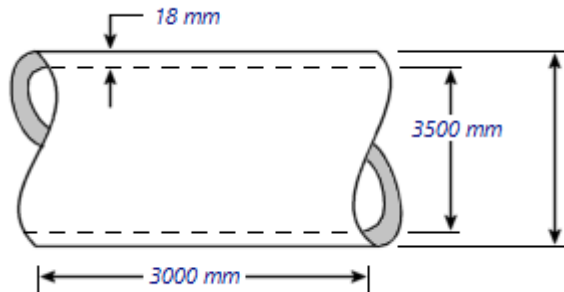
Thickness Ratio = $tr * E^* / (tg - c) = 0.939$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.103 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.939	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	-14 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	1.057 MPa	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3500.000 mm	Actual Stress		124.2 MPa
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		33326016.0 mm²	SG of Contents		0.667
Element Volume		28868.6 ltr	Weight of Contents		50799.6 N
Empty Weight		45359.1 N	Operating Weight		96158.6 N
			Comp. All. Ope. Stress		94.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.057 \cdot 1753.0) / (138.0 \cdot 1.0 - 0.6 \cdot 1.057)$$

$$= 13.4920 + 3.0000 = 16.4920 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (1753.0 + 0.6 \cdot 15.0)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (1750.0 + 0.6 \cdot 18.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.057 \cdot (1753.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 124.187 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.512 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0



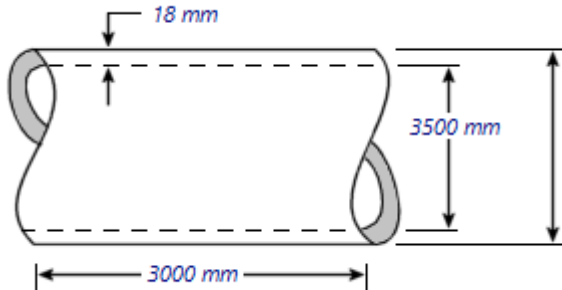
Thickness Ratio = $tr * E^* / (tg - c) = 0.939$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.103 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.939	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	-14 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	1.057 MPa	Int. Design Temperature	200 °C	
Material		SA-516 70	Ext. Design Temperature	200 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3500.000 mm	Actual Stress	124.2 MPa	
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.	18.000 mm	
Surface Area		33326016.0 mm²	SG of Contents	0.667	
Element Volume		28868.6 ltr	Weight of Contents	50799.6 N	
Empty Weight		45359.1 N	Operating Weight	96158.6 N	
			Comp. All. Ope. Stress	94.8 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.057 \cdot 1753.0) / (138.0 \cdot 1.0 - 0.6 \cdot 1.057)$$

$$= 13.4920 + 3.0000 = 16.4920 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (1753.0 + 0.6 \cdot 15.0)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (1750.0 + 0.6 \cdot 18.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.057 \cdot (1753.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 124.187 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.512 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0



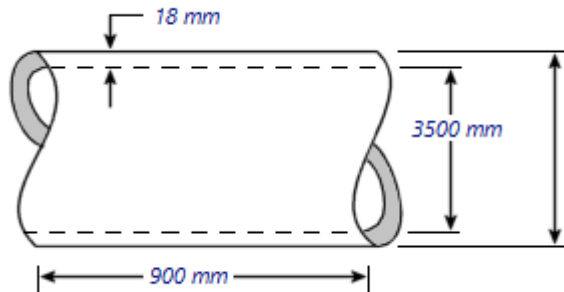
Thickness Ratio = $tr * E^* / (tg - c) = 0.939$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.103 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.939	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	-14 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	1.057 MPa	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		900.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3500.000 mm	Actual Stress		124.2 MPa
Spec. Min. Thk t,e, or sa		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		9997805.0 mm²	SG of Contents		0.667
Element Volume		8660.6 ltr	Weight of Contents		15239.9 N
Empty Weight		13607.7 N	Operating Weight		28847.5 N
			Comp. All. Ope. Stress		94.8 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.057 \cdot 1753.0) / (138.0 \cdot 1.0 - 0.6 \cdot 1.057)$$

$$= 13.4920 + 3.0000 = 16.4920 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (1753.0 + 0.6 \cdot 15.0)$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (1750.0 + 0.6 \cdot 18.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.057 \cdot (1753.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 124.187 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.512 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0



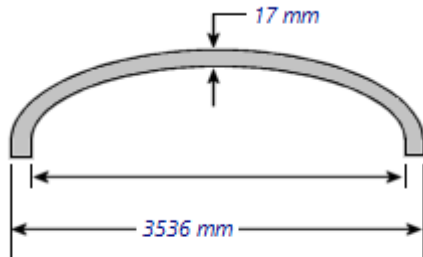
Thickness Ratio = $tr * E^* / (tg - c) = 0.939$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.103 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.939	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	-14 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Right head

Elliptical Head Design Information:					
Design Pressure	P	1.057 MPa	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		200 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Outside Diameter	Do	3536.000 mm	Actual Stress		132.3 MPa
Spec. Min. Thk t,e, or sa		17.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		14155801.0 mm²	SG of Contents		0.667
Element Volume		6104.7 ltr	Weight of Contents		9341.5 N
Empty Weight		23407.8 N	Operating Weight		32749.3 N
			Comp. All. Ope. Stress		94.8 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.950 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.975 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D_o \cdot K_{cor}) / (2 \cdot S \cdot E + 2 \cdot P \cdot (K_{cor} - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (1.057 \cdot 3536.0 \cdot 0.998) / (2 \cdot 138.0 \cdot 1.0 + 2 \cdot 1.057 \cdot (0.998 - 0.1))$$

$$= 13.4217 + 3.0000 = 16.4217 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 14.0) / (0.998 \cdot 3536.0 - 2 \cdot 14.0 \cdot (1.0 - 0.1))$$

$$= 1 - 0.01 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D_o - 2 \cdot t \cdot (K - 0.1)) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 17.0) / (1.0 \cdot 3536.0 - 2 \cdot 17.0 \cdot (1.0 - 0.1))$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D_o - 2 \cdot t \cdot (K_{cor} - 0.1))) / (2 \cdot E \cdot t)$$

$$= (1.057 \cdot (0.998 \cdot 3536.0 - 2 \cdot 14.0 \cdot (0.998 - 0.1))) / (2 \cdot 1.0 \cdot 14.0)$$

$$= 132.258 \text{ MPa}$$



Straight Flange Required Thickness:
 $= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P) + ca$ per Appendix 1-1 (a)(1)
 $= (1.057 \cdot 1768.0) / (138.0 \cdot 1.0 + 0.4 \cdot 1.057) + 3.0$
 $= 16.503 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.007 MPa
 $= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t)$ per Appendix 1-1 (a)(1)
 $= (138.0 \cdot 1.0 \cdot 17.0) / (1768.0 - 0.4 \cdot 17.0)$
 $= 1 - 0.01 = 1 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (3508.0 / (2 \cdot 878.5))^2) / 6$
 $= 0.997725$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / Ro)$ 2.482 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 17.0, tr = 14.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 1.0$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -12 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 20.0, tr = 14.088, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.829$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -17 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1.103 MPa
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.829	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-17 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 43.950 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.975 mm

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.425 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.365 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.706 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.205 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.501 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.501 MPa
App 27-4	= M.A.W.P.	1.096 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 1.096 • 1.0
 = 1.425 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Left head	182.4	234.0	0.780	1.46
Shell-1	171.4	234.0	0.732	1.46
Shell-2	171.4	234.0	0.732	1.46
Shell-3	171.4	234.0	0.732	1.46
Shell-4	171.4	234.0	0.732	1.46
Shell-5	171.4	234.0	0.732	1.46
Right head	182.6	234.0	0.780	1.46

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1 40"	Nozzle	138.00	138.00	1.000
N4 1"	Nozzle	138.00	138.00	1.000
N5A 1"	Nozzle	138.00	138.00	1.000
N5B 1"	Nozzle	138.00	138.00	1.000
N9 8"	Nozzle	118.00	118.00	1.000
N9 8"	Pad	138.00	138.00	1.000
N11 1"	Nozzle	138.00	138.00	1.000
N2 32"	Nozzle	138.00	138.00	1.000
N6A 2"	Nozzle	138.00	138.00	1.000
N6B 2"	Nozzle	138.00	138.00	1.000
N10 1.5"	Nozzle	138.00	138.00	1.000
N12A 1"	Nozzle	138.00	138.00	1.000
N12B 1"	Nozzle	138.00	138.00	1.000
N3 8"	Nozzle	118.00	118.00	1.000
N3 8"	Pad	138.00	138.00	1.000
N8 3"	Nozzle	138.00	138.00	1.000
M1 24"	Nozzle	138.00	138.00	1.000
M1 24"	Pad	138.00	138.00	1.000
N7 3"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
SR1	138.00	138.00	1.000
SR2	138.00	138.00	1.000

Minimum 1.000



Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Left head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Right head	138.00	138.00	1.000
N9 blind flg	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1 40"	26.58	234.00	0.114
N4 1"	3.24	225.00	0.014
N5A 1"	3.24	225.00	0.014
N5B 1"	3.24	225.00	0.014
N9 8"	15.89	216.00	0.074
N11 1"	3.24	225.00	0.014
N2 32"	36.01	234.00	0.154
N6A 2"	4.84	225.00	0.021
N6B 2"	4.84	225.00	0.021
N10 1.5"	3.74	225.00	0.017
N12A 1"	3.24	225.00	0.014
N12B 1"	3.24	225.00	0.014
N3 8"	15.89	216.00	0.074
N8 3"	4.29	225.00	0.019
M1 24"	50.78	234.00	0.217
N7 3"	5.46	225.00	0.024

Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

MDMT Calculations for: SR1, Shell to Ring Junction, Curve: B

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.939$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-14 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations for: SR2, Shell to Ring Junction, Curve: B

Govrn. thk, tg = 14.0, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.939$, Temp. Reduction = 3 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-22 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	3534	14	0.00055021	53.2642
20	Ring	2891.67	3536	15	0.00045542	44.0878
Ring	30	7800	3536	15	0.00016206	15.6888
30	40	7800	3536	15	0.00016206	15.6888
40	50	7800	3536	15	0.00016206	15.6888
50	Ring	7800	3536	15	0.00016206	15.6888
Ring	60	2891.83	3536	15	0.00045539	44.0851
60	70	2891.83	3536	15	0.00045539	44.0851
70	80	No Calc	3536	14	0.0005499	53.2341
80	90	No Calc	...	36.624	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	17	No Calc	...	0.23445
20	Ring	18	No Calc	...	0.24937
Ring	30	18	No Calc	...	0.088737
30	40	18	No Calc	...	0.088737
40	50	18	No Calc	...	0.088737
50	Ring	18	No Calc	...	0.088737
Ring	60	18	No Calc	...	0.24935
60	70	18	No Calc	...	0.24935
70	80	17	No Calc	...	0.23419
80	90	39.624	34.1365	...	No Calc

Minimum

0.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	2891.67	No Calc	No Calc	No Calc
Ring	30	7800	No Calc	No Calc	No Calc
30	40	7800	No Calc	No Calc	No Calc
40	50	7800	No Calc	No Calc	No Calc
50	Ring	7800	No Calc	No Calc	No Calc
Ring	60	2891.83	No Calc	No Calc	No Calc
60	70	2891.83	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	2384.43	6094.47	2026.76	6125.04	166.91
20	30	4625.65	28868.6	3857.98	28967.6	323.795
30	40	4625.65	28868.6	3857.98	28967.6	323.795
40	50	4625.65	28868.6	3857.98	28967.6	323.795
50	60	4625.65	28868.6	3857.98	28967.6	323.795
60	70	1387.69	8660.57	1157.39	8690.29	97.1383
70	80	2387.09	6104.65	2029.03	6135.26	167.096
80	90	35.3451	...	35.3451	...	2.47416
Total		24697	136333.89	20680	136821.06	1728

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Plat	8527.37	6550	1843.2	...	PLATFORM
10	Liqd	951.986	-291.667	1200	...	Liquid 10
10	Insl	99.5626	-412.5	...	...	PP
20	Sadl	1090.27	2550	1888.5	...	Left Saddle
20	Liqd	5180.46	1500	1200	...	Liquid 20
20	Insl	182.464	1500	...	...	Ins: 20
20	Ring	493.19	2550	...	...	SR1
20	Nozl	936.646	1110	2258	...	N1 40"
20	Nozl	8.00157	2250	1762.7	...	N4 1"
20	Nozl	8.00157	950	-498.501	1690.74	N5A 1"
20	Nozl	4.8497	950	-1762.7	...	N5B 1"
20	Nozl	138.576	2750	1846.84	...	N9 8"
20	Nozl	8.00157	2400	1762.7	...	N11 1"
20	Wght	500	1150	1750	...	N1 Insert plate
20	Wght	25	2550	...	...	Fy wt-1
20	Wght	230	1000	875	1515.54	Lifting lug 1
20	Wght	230	1000	875	-1515.54	Lifting lug 2
20	Forc	...	2550	...	...	Nozzles F/M 1
30	Liqd	5180.46	1500	1200	...	Liquid 30
30	Insl	182.464	1500	...	...	Ins: 30
30	Wght	1150	510	...	...	Cleat weight
40	Liqd	5180.46	1500	1200	...	Liquid 40
40	Insl	182.464	1500	...	...	Ins: 40
50	Sadl	1090.27	1350	1888.5	...	22
50	Liqd	5180.46	1500	1200	...	Liquid 50
50	Insl	182.464	1500	...	...	Ins: 50
50	Ring	345.233	1350	...	...	SR2
50	Nozl	552.031	2990	2156.4	...	N2 32"
50	Nozl	8.52166	2650	-502.092	1702.92	N6A 2"
50	Nozl	8.52166	2650	-1775.4	...	N6B 2"
50	Nozl	7.52987	2950	-1525.91	895.068	N10 1.5"
50	Nozl	4.84996	2350	-498.501	1690.74	N12A 1"
50	Nozl	4.84996	2350	-1762.7	...	N12B 1"
50	Wght	232	2990	1750	...	N2 insert pleat
50	Wght	25	1350	...	...	Fy wt-2
50	Wght	230	2900	875	1515.54	Lifting lug 3
50	Wght	230	2900	875	-1515.54	Lifting lug 4
50	Forc	...	1350	...	...	Nozzle F/M 2
60	Liqd	1554.14	450	1200	...	Liquid 60
60	Insl	54.7392	450	...	...	Ins: 60
60	Nozl	63.2948	310	-1846.84	...	N3 8"
60	Nozl	20.0361	720	-1788.1	...	N8 3"
60	Wght	10	310	-1600	...	N3 Vortex break
70	Liqd	952.631	341.833	1201	...	Liquid 70
70	Insl	99.6728	462.5	...	...	Ins: 70
70	Nozl	454.384	925.5	...	...	M1 24"



70 Wght	50	925.5	...	...	M1 davit
80 Insl	0.79972	...	19.812	...	Ins: 80
80 Nozl	15.2071	...	164.63	...	N7 3"

Total Weight of Each Detail Type:

Saddles	2180.5
Platforms	8527.4
Liquid	24180.6
Insulation	984.6
Stiffeners	838.4
Nozzles	2243.3
Weights	2912.0
Sum of the Detail Weights	41866.9 kgm

Note:

Because some nozzles were attached to a head or flange, their weights include the attached head or flange.

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	26388.1	26388.1	26388.1	26388.1	26388.1	26388.1
Saddles	2180.5	2180.5	2180.5	2180.5	2180.5	2180.5
Stif. Rings	838.4	838.4	838.4	838.4	838.4	838.4
Nozzles	2243.3	2243.3	2243.3	2243.3	2243.3	2243.3
Wld Weights	2862.0	2862.0	2862.0	2862.0	2862.0	2862.0
Platforms	...	...	...	8527.4	8527.4	8527.4
Insulation	...	...	...	984.6	984.6	984.6
Empty Weights	...	...	...	50.0	50.0	...
Test Weights	...	50.0	...	...	...	...
Ope Weights	...	...	...	...	...	50.0
Ope. Liquid	...	...	...	...	...	24180.6
Test Liquid	...	136250.7	...	...	...	...
Totals	34512.4	170813.1	34512.4	44074.4	44074.4	68255.0

Field Installation Options:

Platforms installed before lifting.
 Insulation installed before lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	34512.4 kgm
Shop Test Wt.	- Fabricated weight + water (full)	170813.1 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	34512.4 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	44074.4 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	44074.4 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	68255.0 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	63957.1 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
------	----	---	--



10	20	14140132	13828598
20	30	33326016	32986722
30	40	33326016	32986722
40	50	33326016	32986722
50	60	33326016	32986722
60	70	9997805	9896017
70	80	14155801	13844095
80	90	161887	...

Total	171759696.0	169515600.0 mm ²
Total	171.8	169.5 m ²

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1 40"	4.38	5.10	200	300	GR 1.1	0.921	3.898	4.380	4.380
N4 1"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N5A 1"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N5B 1"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N9 8"	4.38	5.10	200	300	GR 1.1	2.764	3.806	4.380	3.806
N11 1"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N2 32"	4.38	5.10	200	300	GR 1.1	1.004	3.814	4.380	4.380
N6A 2"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N6B 2"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N10 1.5"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N12A 1"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N12B 1"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N3 8"	4.38	5.10	200	300	GR 1.1	2.764	3.806	4.380	3.806
N8 3"	4.38	5.10	200	300	GR 1.1	4.219	2.351	4.380	2.351
M1 24"	1.38	1.96	200	150	GR 1.1	...	...	...	...
N7 3"	1.38	1.96	200	150	GR 1.1	...	...	...	...
Min Rating	1.380	1.960, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1 40"	1046.226	0.10	200000000	27000	1.096	3.898	Pass
N9 8"	248.539	0.50	8000000	5400	1.096	3.806	Pass
N2 32"	876.300	0.10	127999936	21600	1.096	3.814	Pass
N3 8"	248.539	0.50	8000000	5400	1.096	3.806	Pass
N8 3"	111.125	0.50	1081665	2000	1.096	2.351	Pass

Evaluating Flange on Nozzle [N1 40"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(200000000) + 4 \cdot \text{abs}(27000)1046.226 \leq \pi \cdot 1046.226^3 \cdot ((4.380 - 1.096) + 1.00 \cdot 4.380)$$

$$= 0.33130E+10 \leq 0.13397E+11 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1 40" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.380, 0.100 \cdot 4.380 + 4.380 - 3313038080.0 / (\pi \cdot 1046.226^3))$$

$$= \min(4.380, 3.898)$$

$$= 3.898 \text{ MPa}$$

Evaluating Flange on Nozzle [N9 8"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(8000000) + 4 \cdot \text{abs}(5400)248.539 \leq \pi \cdot 248.539^3 \cdot ((4.380 - 1.096) + 1.00 \cdot 4.380)$$

$$= 0.13337E+09 \leq 0.26413E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N9 8" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.380, 0.500 \cdot 4.380 + 4.380 - 133370528.0 / (\pi \cdot 248.539^3))$$

$$= \min(4.380, 3.806)$$

$$= 3.806 \text{ MPa}$$

Evaluating Flange on Nozzle [N2 32"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(127999936) + 4 \cdot \text{abs}(21600)876.300 \leq \pi \cdot 876.300^3 [(4.380 - 1.096) + 1.100 \cdot 4.380] \\ &= 0.21237E+10 \leq 0.78718E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2 32" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.380, 0.100 \cdot 4.380 + 4.380 - 2123742080.0 / (\pi \cdot 876.300^3)) \\ &= \min(4.380, 3.814) \\ &= 3.814 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N3 8"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(8000000) + 4 \cdot \text{abs}(5400)248.539 \leq \pi \cdot 248.539^3 [(4.380 - 1.096) + 1.500 \cdot 4.380] \\ &= 0.13337E+09 \leq 0.26413E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3 8" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.380, 0.500 \cdot 4.380 + 4.380 - 133370528.0 / (\pi \cdot 248.539^3)) \\ &= \min(4.380, 3.806) \\ &= 3.806 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N8 3"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1081665) + 4 \cdot \text{abs}(2000)111.125 \leq \pi \cdot 111.125^3 [(4.380 - 1.096) + 1.500 \cdot 4.380] \\ &= 0.18196E+08 \leq 0.23609E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N8 3" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.380, 0.500 \cdot 4.380 + 4.380 - 18196008.0 / (\pi \cdot 111.125^3)) \\ &= \min(4.380, 2.351) \\ &= 2.351 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 0.169

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 0.83 • 1.0 • 0.6 • 57.5²)

= 1653.9 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 2.9 • 0.7 • 1653.942

= 3339.4 N

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	2365.00	1.0000	1.0000	1.0000	57.50	0.7000
20	2365.00	1.0000	1.0000	1.0000	57.50	0.7000
30	2365.00	1.0000	1.0000	1.0000	57.50	0.7000
40	2365.00	1.0000	1.0000	1.0000	57.50	0.7000
50	2365.00	1.0000	1.0000	1.0000	57.50	0.7000
60	2365.00	1.0000	1.0000	1.0000	57.50	0.7000
70	2365.00	1.0000	1.0000	1.0000	57.50	0.7000
80	2280.19	1.0000	1.0000	1.0000	57.50	0.7000

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLATFORM	17040000.00	2365.00	1.65	28193.33	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
N1 Insert pla	0.00	4115.00	1.65	0.00
Fy wt-1	0.00	2365.00	1.65	0.00
Lifting lug 1	0.00	3240.00	1.65	0.00



Lifting lug 2	0.00	3240.00	1.65	0.00
Cleat weight	0.00	2365.00	1.65	0.00
N2 insert ple	0.00	4115.00	1.65	0.00
Fy wt-2	0.00	2365.00	1.65	0.00
Lifting lug 3	0.00	3240.00	1.65	0.00
Lifting lug 4	0.00	3240.00	1.65	0.00
N3 Vortex bre	0.00	300.00	1.65	0.00
M1 davit	0.00	2365.00	1.65	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	2365	4348.8	2884407	1.65458	34686.1
20	30	2365	4351.2	13053601	1.65458	16624.2
30	40	2365	4351.2	13053601	1.65458	16624.2
40	50	2365	4351.2	13053601	1.65458	16624.2
50	60	2365	4351.2	13053601	1.65458	16624.2
60	70	2365	4351.2	3916080	1.65458	4987.26
70	80	2365	4351.2	2887423	1.65458	3677.23
80	90	2280.19	565.2	22395.5	1.65458	28.5214

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2015 + 2018 Amendment 2

Input Data:

Importance Factor as Entered by User I 1.5000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 1.5000
 Time Period of Vibration (program or user) 0.0303 Seconds
 Value of Sa/g from user Spectrum Data 0.1023
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa / g \cdot DampVal) / (R / I)$
 $= (0.1023 \cdot 1.4) / 1.0$
 $= 0.143$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.081
0.050	0.123
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.104
0.800	0.088
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
 Computed Sa/g = 0.1023

Note:

It was indicated to apply the damping factor, this assumes that the Site Spectrum specified is for 5% damping. If your Sa/g values are already per the actual damping value of the vessel then do not apply the damping factor.

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 67308.99 \cdot 0.1432 \cdot 1.0$
 $= 9641.7 \text{ N}$

Vertical load per 7.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.1432 \cdot 605781 \cdot 1.0$
 $= 57850.2 \text{ N}$

Earthquake Load Calculation:

	Earthquake	Earthquake	Element
--	------------	------------	---------



From	To	Height mm	Weight N	Ope Load N
10	20	1750	67309	9641.71
20	30	1750	67309	9641.71
20	30	1750	67309	9641.71
30	40	1750	67309	9641.71
40	50	1750	67309	9641.71
50	0	1750	67309	9641.71
50	60	1750	67309	9641.71
60	70	1750	67309	9641.71
70	80	1751	67309	9641.71

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
 Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	6500.000 mm
Center of Gravity of the Platforms	6550.000 mm
Center of Gravity of the Liquid	6500.188 mm
Center of Gravity of the Insulation	6497.768 mm
Center of Gravity of the Stiffening Rings	5811.764 mm
Center of Gravity of the Nozzles	7113.507 mm
Center of Gravity of the Added Weights (Operating)	5017.051 mm
Center of Gravity of the Added Weights (Empty)	5017.051 mm
Center of Gravity of Bare Shell New and Cold	6495.452 mm
Center of Gravity of Bare Shell Corroded	6494.430 mm
Vessel CG in the Operating Condition	6449.782 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	6394.819 mm
Vessel CG in the Test Condition	6479.122 mm

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	13000.00 mm
Total vessel weight (No Liquid) Twt	432193.34 N
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	648290.00 N
Distance to the Lug on the Left End	1000.00 mm
Distance to the Lug on the Right End	12000.00 mm
Design Reaction force at the Left End Lug	330343.91 N
Design Reaction force at the Right End Lug	317946.09 N
CG distance during field lifting operation	6394.82 mm
CG distance from the left end lug/trunnion	5394.82 mm
CG distance from the nearer lifting lug/trunnion	5394.82 mm
Compressive Allowable Increase Factor	1.00

Critical Values: (MPa, mm)

	Max Stress	Elevation	Allowables
Bending	1.81	6650.00	92.37 (UG-23)
Shear	1.04	11450.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment N-mm	Bending Stress MPa	Shear Force N	Shear Stress MPa
0.00	0.0	0.0	124855.8	0.7
650.00	97066776.0	0.6	163273.0	0.8
4250.00	235579824.0	1.4	23704.0	0.1
7850.00	303654112.0	1.7	62734.7	0.3
11450.00	8755863.0	0.1	206799.2	1.0
12950.00	26940696.0	0.2	9604.3	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0196 at elevation 6650.0 mm - Must be <=1



Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

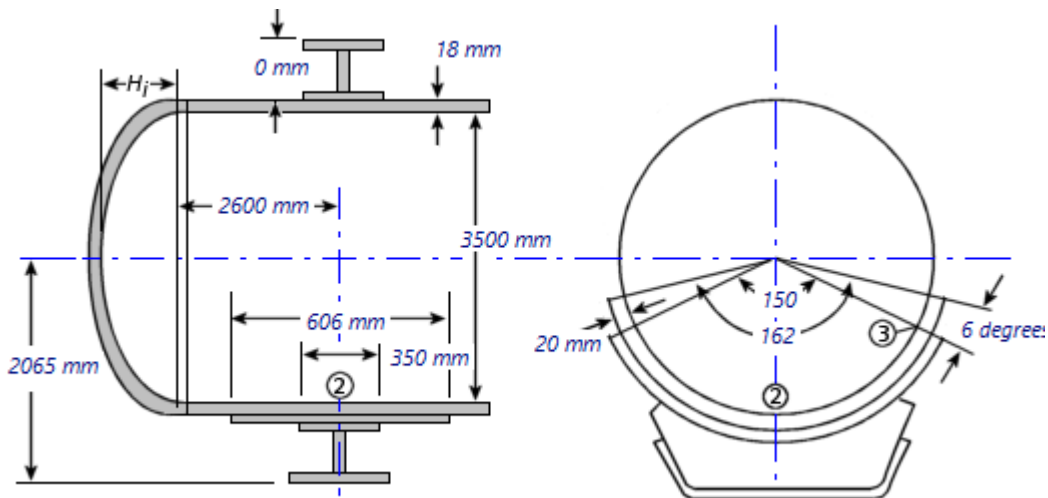
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1760.50	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	13000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2600.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	606.00	mm
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	885.00	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2065.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	636039.69	N
Pressure used in Analysis	P	1.050 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	59.61	138.00
Long. Stress at Bottom of Midspan	63.63	138.00
Long. Stress at Top of Saddles	64.28	138.00
Long. Stress at Bottom of Saddles	58.95	138.00
Tangential Shear in Shell	4.22	110.40
Circ. Compressive Stress in Shell	4.70	138.00
Stiffener Circ. Stress at Shell	22.24	172.50
Stiffener Circ. Stress at Tip	120.86	172.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(109875.9/2 + 110090) \cdot 2065.0/3400.0002$
 $= 300690.6 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind; Sum of X Forces}) \cdot B / Ls$
 $= \max(34785.68, 251950) \cdot 2065.0/7800.0$
 $= 66702.1 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq; Sum of X Forces}) \cdot B / Ls$
 $= \max(86775.37, 251950) \cdot 2065.0/7800.0$
 $= 66702.1 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(86775/2 + 110090) \cdot 2065.0/3400.0002$
 $= 279645.3 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$
 $= 335349 + \max(66702, 300691, 66702, 279645)$
 $= 636039.7 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1654.611(4.243 + \pi/4(3.626 \cdot 1.2)^2)$
 $= 34787.480 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	646730.88	N
Transverse Shear Load	165027.95	N
Longitudinal Shear Load	251950.00	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 3400.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0252
K7p = 0.0252			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -636040 \cdot 2600.0 [1 - (1 - 2600.0/13000.0 + (1760.5^2 - 885.0^2)/(2 \cdot 2600.0 \cdot 13000.0))/(1 + (4 \cdot 885.0)/(3 \cdot 13000.0))]$
 $= -389044864.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m)/3L) - 4a/L)$
 $= 636040 \cdot 13000/4((1 + 2(1760^2 - 885^2)/(13000^2))/(1 + (4 \cdot 885)/(3 \cdot 13000)) - 4 \cdot 2600/13000)$
 $= 293471776.0 \text{ N-mm}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$
 $= 1.05 \cdot 1760.5/(2 \cdot 15.0) - 0.29347E+09/(\pi \cdot 1760.5^2 \cdot 15.0)$
 $= 59.61 \text{ MPa}$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 1.05 \cdot 1760.5 / (2 \cdot 15.0) + 0.29347E+09 / (\pi \cdot 1760.5^2 \cdot 15.0)$
 $= 63.63 \text{ MPa}$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:
 $= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$
 $= 1.05 \cdot 1760.5 / (2 \cdot 15.0) - 0.4E+09 / (\pi \cdot 1760.5^2 \cdot 15.0)$
 $= 64.28 \text{ MPa}$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:
 $= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$
 $= 1.05 \cdot 1760.5 / (2 \cdot 15.0) + 0.4E+09 / (\pi \cdot 1760.5^2 \cdot 15.0)$
 $= 58.95 \text{ MPa}$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a) / (L + (4 \cdot h_m / 3))$
 $= 636040(13000.0 - 2 \cdot 2600.0) / (13000.0 + (4 \cdot 885.0 / 3))$
 $= 349866.7 \text{ N}$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:
 $= T / (\pi \cdot R_m \cdot t)$
 $= 349866.69 / (\pi \cdot 1760.4999 \cdot 15.0)$
 $= 4.22 \text{ MPa}$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(1760.5 \cdot 15.0)}$
 $= 126.753 \text{ mm}$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:
 $= -K_5 \cdot Q \cdot k/A$
 $= -0.673 \cdot 636040 \cdot 0.1/9102.59$
 $= -4.70 \text{ MPa}$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:
 $= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$
 $= -0.3021 \cdot 636040/9102.59 + 0.0317 \cdot 636040 \cdot 1760.5 \cdot 80.28/65682084$
 $= 22.24 \text{ MPa}$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:
 $= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$
 $= -0.3021 \cdot 636040/9102.59 - 0.0317 \cdot 636040 \cdot 1760.5 \cdot 184.72/65682084$
 $= -120.86 \text{ MPa}$

Distance between Saddle Supports [Ls]:
 $= 7800.0 \text{ mm}$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:
 $= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$
 $= 0.000013 \cdot 7800.0(200.0 - 21.1)$
 $= 17.790 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3400.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	225.0	MPa
Height of Web at Center	Hw,c	259.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1753.0 \cdot 15.0)} + 606.0)^3) / 12$$

$$= 241583.797 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 241583.797 + 12884.48 (88.832 - 7.5)^2$$

$$= 85470048.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	859.0	15.0	7.5	12884.5	96633.6	0.855E+08
Wearplate	606.0	20.0	25.0	12120.0	303000.0	0.498E+08
Web	10.0	255.0	162.5	2550.0	414375.4	0.277E+08
BasePlate	350.0	22.0	301.0	7700.0	2317701.2	0.347E+09
Totals	...	...	...	35254.5	3131710.2	0.510E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3131710 / 35254.48$$

$$= 88.832 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 636040$$

$$= 164971.1094 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 7.3753 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 134.9988 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2065.0 - 1753.0 \cdot \sin(150.0/2) / 1.309$$

$$= 771.439 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 127316704.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 22.1757 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 149.9987 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(636040 + 10691) \cdot 350.0) / (4 \cdot 3400.0 \cdot 150.0)}$$

$$= 18.246 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (1750.0 + 18.0 + 20.0)$$

$$= 3454.151 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 3454.1506 / (5 - 1)$$

$$= 863.538 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot Bpwid / 2 \\
 &= 863.5377 \cdot 350.0 / 2 \\
 &= 151119.094 \text{ mm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth) \\
 &= 646731 / (3400.0 \cdot 350.0) \\
 &= 0.543 \text{ N/mm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 151119.1 \cdot 0.54 \\
 &= 82128.898 \text{ N}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= Rib \text{ Area} + Web \text{ Area} \\
 &= 3400.0 + 4317.688 \\
 &= 7717.688 \text{ mm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / Ar \\
 &= 82128.9 / 7717.688 \\
 &= 10.643 \text{ MPa}
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	3500.0	...	0.357E+08

Rib dimension [D]:

$$\begin{aligned}
 &= Saddle \text{ Width} - Web \text{ Thickness} \\
 &= 350.0 - 10.0 \\
 &= 340.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0 / 7717.688 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= Saddle \text{ Width} / 2 \\
 &= 350.0 / 2 \\
 &= 175.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) } \\
 &= \sqrt{ (35729128 / 7717.688) } \\
 &= 68.041 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned}
 &= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk \\
 &= 2065.0 - \cos(150.0 / 2)(1750.0 + 18.0 + 20.0) - 22.0 \\
 &= 1580.232 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned}
 &= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) \\
 &= 2 \cdot \sin(150.0 / 2)(1750.0 + 18.0 + 20.0) \\
 &= 3454.151 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned}
 &= Baseplate \text{ length} - clearance \\
 &= 3400.0 - 50.8 \\
 &= 3349.200 \text{ mm}
 \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned}
 &= (Lw - Bd) / 2 \\
 &= (3454.151 - 3349.2) / 2
 \end{aligned}$$

= 52.475 mm

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1580.232^2 + 52.475^2}$$

$$= 1581.103 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199955 / 225.0}$$

$$= 132.447$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1581.103 / 68.041$$

$$= 23.238$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 251950.0 / (2 \cdot 3400.0) \cdot 863.538 \cdot 1581.1 / 2$$

$$= 25304212.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (23.2377 < 132.4469) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.24)^2 / (2 \cdot 132.45^2)) 225 /$$

$$(5/3 + 3 \cdot (23.24) / (8 \cdot 132.45) - (23.24^3) / (8 \cdot 132.45^3))$$

$$= 127.9 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 10.64 / 127.92 + (25304212 \cdot 175.0 / 35729164) / 150.0$$

$$= 0.909$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	3400.0	0.0	0.357E+08
Web	863.5	10.0	0.0	8635.4	0.0	0.720E+05
Totals	...	...	...	12035.4	...	0.358E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 12035.376$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350.0 / 2$$

$$= 175.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2065.0 - \cos(1788.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 255.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{35800256 / 12035.376}$$

$$= 54.540 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 255.0 / 54.54$$

$$= 4.675$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 251950.0 / (2 \cdot 3400.0)$$

= 37.051 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 37.051 • 863.538 • 255.0
 = 8162124.500 N-mm

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 8162125/204573.094
 = 39.886 MPa

Compressive Allowable, $KL/r < C_c$ (4.6755 < 132.4469) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)
 = (1-(4.68)²/(2 • 132.45²))225/
 (5/3+3•(4.68)/(8• 132.45)-(4.68³)/(8•132.45³)
 = 133.9 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$
 = 13.94/133.85 + (8162125 • 175.0/35800256)/150.0
 = 0.370

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.30 MPa
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	738.0151 mm ²
Saddle Load QO (Weight)	QO	346040.2 N
Saddle Load QL (Wind/Seismic contribution)	QL	66702.1 N
Maximum Transverse Force	Ft	165028.0 N
Maximum Longitudinal Force	Fl	251950.0 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

= 33934.4 N

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= max(Ff, Fl) / (Bolt Area • Number of Bolts)
 = 251950.0/(738.015 • 4)
 = 85.4 MPa. Must be less than 107.9 MPa.

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)
 = 165028.0/(738.015 • 4)
 = 55.9 MPa. Must be less than 107.9 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= Fl / (BoltShearAllowable • Nbolts)
 = 251950.0/(107.87 • 4.0)
 = 583.9539 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments
 = 2065.0 • 165027.95 + 0.0

$$= 340920928.00 \text{ N-mm}$$

Eccentricity (e):
 $= R_{mom} / QO$
 $= 0.34092E+09/346040.25$
 $= 984.81 \text{ mm} > B_{plen}/6 \rightarrow \text{Uplift in Transverse direction}$

$$f = B_{plen} / 2 - Edgedis$$

$$= 3400.0/2 - 50.8$$

$$= 1649.20 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):
 $= E_S / E_C$
 $= 203402.5/21526.32$
 $= 9.45$

$$K1 = 3 (e - 0.5 \cdot B_{plen})$$

$$= 3 (984.81 - 0.5 \cdot 3400.0)$$

$$= -2145.58 \text{ mm}$$

$$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$$

$$= 6 \cdot 9.45 \cdot 1476.03/350.0 \cdot (1649.2 + 984.81)$$

$$= 629769.77 \text{ mm}^2$$

$$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$$

$$= -629769.75 \cdot (0.5 \cdot 3400.0 + 1649.2)$$

$$= -2109224957.98 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$$

$$Y^3 + -2145.58 \cdot Y^2 + 629770 \cdot Y + -0.2E+10 = 0$$

$$Y = 2276.04 \text{ mm}$$

$$Num = (B_{plen} / 2 - Y / 3 - e)$$

$$= (3400.0/2 - 2276.04/3 - 984.81)$$

$$= -43.49$$

$$Denom = (B_{plen} / 2 - Y / 3 + f)$$

$$= (3400.0/2 - 2276.04/3 + 1649.2)$$

$$= 2590.52$$

Total Bolt Tension Force [Tforce]:

$$= -QO \cdot Num / Denom$$

$$= -346040.25 \cdot -43.49/2590.52$$

$$= 5809.02 \text{ N}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$= Tforce / (Stba \cdot Nbt)$$

$$= 5809.02 / (137.3 \cdot 2.0)$$

$$= 21.1563 \text{ mm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 583.9539, 21.1563]$$

$$= 583.9539 \text{ mm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + Tforce) / (Y \cdot B_{pwid})$$

$$= 2(346040.25 + 5809.02) / (2276.04 \cdot 350.0)$$

$$= 0.88 \text{ MPa}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (B_{plen} - WebLength) / 2$$

$$= (3400.0 - 3349.2) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:



$$= T_{force} \cdot A_{dist}$$

$$= 5809.02 \cdot 25.4$$

$$= 147608.94 \text{ N-mm}$$

Equivalent Bearing Pressure (f1):

$$= f_c \cdot (Y - A_{dist}) / Y$$

$$= 0.88 \cdot (2276.04 - 25.4) / 2276.04$$

$$= 0.87 \text{ MPa}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (A_{dist}^2 \cdot B_{pwid} / 6) \cdot (f1 + 2 \cdot f_c)$$

$$= (25.4^2 \cdot 350.0 / 6) \cdot (0.87 + 2 \cdot 0.88)$$

$$= 99403.40 \text{ N-mm}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(M_t, M_c) / (B_{pwid} \cdot S_{ba}))}$$

$$= \sqrt{(6 \cdot \max(147608.94, 99403.4) / (350.0 \cdot 207.0))}$$

$$= 3.4958 \text{ mm}$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.939, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Input and Calculated Values:

Vessel Mean Radius	Rm	1760.50	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	13000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	10400.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	606.00	mm
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	885.50	mm
Shell Allowable Stress used in Calculation		138.00	MPa
Head Allowable Stress used in Calculation		138.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2065.00	mm
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 629343.62 N
 Pressure used in Analysis P 1.050 MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top of Midspan		93.24	138.00
Long. Stress at Bottom of Midspan		29.99	138.00
Long. Stress at Top of Saddles		97.87	138.00
Long. Stress at Bottom of Saddles		25.37	138.00

Tangential Shear in Shell	4.17	110.40
Circ. Compressive Stress in Shell	5.64	138.00
Stiffener Circ. Stress at Shell	22.22	172.50
Stiffener Circ. Stress at Tip	159.71	172.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(109875.9/2 + 110090) \cdot 2065.0/3400.0002$
 $= 300690.6 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(34785.68, 251950) \cdot 2065.0/7800.0$
 $= 66702.1 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(86775.37, 251950) \cdot 2065.0/7800.0$
 $= 66702.1 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(86775/2 + 110090) \cdot 2065.0/3400.0002$
 $= 279645.3 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 328653 + \max(66702, 300691, 66702, 279645)$
 $= 629343.6 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1654.611(4.243 + \pi/4(3.626 \cdot 1.2)^2)$
 $= 34787.480 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	640034.81	N
Transverse Shear Load	165027.95	N
Longitudinal Shear Load	251950.00	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 3400.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0252
K7p = 0.0252			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -629344 \cdot 10400.0 [1 - (1 - 10400.0/13000.0 + (1760.5^2 - 885.5^2)/(2 \cdot 10400.0 \cdot 13000.0))/(1 + (4 \cdot 885.5)/(3 \cdot 13000.0))]$
 $= -5295898112.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m)/(3L)) - 4a/L)$
 $= 629344 \cdot 13000/4((1 + 2(1760^2 - 886^2)/(13000^2))/(1 + (4 \cdot 886)/(3 \cdot 13000)) - 4 \cdot 10400/13000)$

= -4620598784.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.05 \cdot 1760.5 / (2 \cdot 15.0) - 0.5E+10 / (\pi \cdot 1760.5^2 \cdot 15.0)$$

$$= 93.24 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.05 \cdot 1760.5 / (2 \cdot 15.0) + 0.5E+10 / (\pi \cdot 1760.5^2 \cdot 15.0)$$

$$= 29.99 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.05 \cdot 1760.5 / (2 \cdot 15.0) - 0.5E+10 / (\pi \cdot 1760.5^2 \cdot 15.0)$$

$$= 97.87 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1.05 \cdot 1760.5 / (2 \cdot 15.0) + 0.5E+10 / (\pi \cdot 1760.5^2 \cdot 15.0)$$

$$= 25.37 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 629344(13000.0 - 2 \cdot 10400.0) / (13000.0 + (4 \cdot 885.5 / 3))$$

$$= -346167.1 \text{ N}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= -346167.06 / (\pi \cdot 1760.4999 \cdot 15.0)$$

$$= -4.17 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1760.5 \cdot 15.0)}$$

$$= 126.753 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.673 \cdot 629344 \cdot 0.1 / 7512.59$$

$$= -5.64 \text{ MPa}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.3021 \cdot 629344 / 7512.59 + 0.0317 \cdot 629344 \cdot 1760.5 \cdot 69.23 / 51124100$$

$$= 22.22 \text{ MPa}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.3021 \cdot 629344 / 7512.59 - 0.0317 \cdot 629344 \cdot 1760.5 \cdot 195.77 / 51124100$$

$$= -159.71 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3400.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	225.0	MPa
Height of Web at Center	Hw,c	259.0	mm
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1753.0 \cdot 15.0)} + 606.0)^3) / 12$$

$$= 241583.797 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 241583.797 + 12884.48 (88.832 - 7.5)^2$$

$$= 85470048.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	859.0	15.0	7.5	12884.5	96633.6	0.855E+08
Wearplate	606.0	20.0	25.0	12120.0	303000.0	0.498E+08
Web	10.0	255.0	162.5	2550.0	414375.4	0.277E+08
BasePlate	350.0	22.0	301.0	7700.0	2317701.2	0.347E+09
Totals	...	...	...	35254.5	3131710.2	0.510E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3131710 / 35254.48$$

$$= 88.832 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 629344$$

$$= 163234.3281 \text{ N}$$

$$\text{Tension Stress, } S_t = (F_h / A_s) = 7.2976 \text{ MPa}$$

$$\text{Allowed Stress, } S_a = 0.6 \cdot \text{Yield Str} = 134.9988 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2065.0 - 1753.0 \cdot \sin(150.0/2) / 1.309$$

$$= 771.439 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 125976336.0000 \text{ N-mm}$$

$$\text{Bending Stress, } S_b = (M \cdot C1 / I) = 21.9422 \text{ MPa}$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 149.9987 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(629344 + 10691) \cdot 350.0) / (4 \cdot 3400.0 \cdot 150.0)}$$

$$= 18.151 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (1750.0 + 18.0 + 20.0)$$

$$= 3454.151 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 3454.1506 / (5 - 1)$$

$$= 863.538 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 863.5377 \cdot 350.0 / 2$$

$$= 151119.094 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 640035 / (3400.0 \cdot 350.0)$$

$$= 0.538 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 151119.1 \cdot 0.54$$

$$= 81278.555 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 3400.0 + 4317.688$$

$$= 7717.688 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 81278.6 / 7717.688$$

$$= 10.532 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	3500.0	...	0.357E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 350.0 - 10.0$$

$$= 340.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 7717.688$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350.0 / 2$$

$$= 175.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(35729128 / 7717.688)}$$

$$= 68.041 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2065.0 - \cos(150.0 / 2) (1750.0 + 18.0 + 20.0) - 22.0$$

$$= 1580.232 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(150.0 / 2) (1750.0 + 18.0 + 20.0)$$

$$= 3454.151 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 3400.0 - 50.8$$

$$= 3349.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (3454.151 - 3349.2) / 2$$

= 52.475 mm

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1580.232^2 + 52.475^2}$$

$$= 1581.103 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 199955 / 225.0}$$

$$= 132.447$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1581.103 / 68.041$$

$$= 23.238$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 251950.0 / (2 \cdot 3400.0) \cdot 863.538 \cdot 1581.1 / 2$$

$$= 25304212.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (23.2377 < 132.4469) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.24)^2 / (2 \cdot 132.45^2)) 225 /$$

$$(5/3 + 3 \cdot (23.24) / (8 \cdot 132.45) - (23.24^3) / (8 \cdot 132.45^3))$$

$$= 127.9 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 10.53 / 127.92 + (25304212 \cdot 175.0 / 35729164) / 150.0$$

$$= 0.908$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	3400.0	0.0	0.357E+08
Web	863.5	10.0	0.0	8635.4	0.0	0.720E+05
Totals	...	...	...	12035.4	...	0.358E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 12035.376$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350.0 / 2$$

$$= 175.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2065.0 - \cos(1788.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 255.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{35800256 / 12035.376}$$

$$= 54.540 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 255.0 / 54.54$$

$$= 4.675$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 251950.0 / (2 \cdot 3400.0)$$

= 37.051 N/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 37.051 • 863.538 • 255.0
 = 8162124.500 N-mm

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 8162125/204573.094
 = 39.886 MPa

Compressive Allowable, $KL/r < C_c$ (4.6755 < 132.4469) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)
 = (1-(4.68)²/(2 • 132.45²))225/
 (5/3+3•(4.68)/(8• 132.45)-(4.68³)/(8•132.45³))
 = 133.9 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$
 = 13.8/133.85 + (8162125 • 175.0/35800256)/150.0
 = 0.369

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.30 MPa
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BlArea	738.0151 mm ²
Saddle Load QO (Weight)	QO	339344.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	66702.1 N
Maximum Transverse Force	Ft	165028.0 N
Maximum Longitudinal Force	Fl	251950.0 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 251950.0/(738.015 • 4)
 = 85.4 MPa. Must be less than 107.9 MPa.

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 165028.0/(738.015 • 4)
 = 55.9 MPa. Must be less than 107.9 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = 251950.0/(107.87 • 4.0)
 = 583.9539 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$
 = 2065.0 • 165027.95 + 0.0
 = 340920928.00 N-mm

Eccentricity (e):



$$\begin{aligned}
 &= R_{mom} / QO \\
 &= 0.34092E+09/339344.12 \\
 &= 1004.24 \text{ mm} > B_{plen}/6 \rightarrow \text{Uplift in Transverse direction}
 \end{aligned}$$

$$\begin{aligned}
 f &= B_{plen} / 2 - e_{gedis} \\
 &= 3400.0/2 - 50.8 \\
 &= 1649.20 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Modular Ratio Of Steel/Concrete (n1):} \\
 &= E_S / E_C \\
 &= 203402.5/21526.32 \\
 &= 9.45
 \end{aligned}$$

$$\begin{aligned}
 K1 &= 3 (e - 0.5 \cdot B_{plen}) \\
 &= 3 (1004.24 - 0.5 \cdot 3400.0) \\
 &= -2087.28 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 K2 &= 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e) \\
 &= 6 \cdot 9.45 \cdot 1476.03/350.0 \cdot (1649.2 + 1004.24) \\
 &= 634416.00 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 K3 &= -K2 \cdot (0.5 \cdot B_{plen} + f) \\
 &= -634416.0 \cdot (0.5 \cdot 3400.0 + 1649.2) \\
 &= -2124786087.87 \text{ mm}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{Iteratively Solving for the Effective Bearing Length:} \\
 Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\
 Y^3 + -2087.28 \cdot Y^2 + 634416 \cdot Y + -0.2E+10 &= 0 \\
 Y &= 2230.05 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Num} &= (B_{plen} / 2 - Y / 3 - e) \\
 &= (3400.0/2 - 2230.05/3 - 1004.24) \\
 &= -47.59
 \end{aligned}$$

$$\begin{aligned}
 \text{Denom} &= (B_{plen} / 2 - Y / 3 + f) \\
 &= (3400.0/2 - 2230.05/3 + 1649.2) \\
 &= 2605.85
 \end{aligned}$$

$$\begin{aligned}
 \text{Total Bolt Tension Force [Tforce]:} \\
 &= - QO \cdot \text{Num} / \text{Denom} \\
 &= - 339344.12 \cdot -47.59/2605.85 \\
 &= 6197.31 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Bolt Area Required due to Transverse Load [Bltareart]:} \\
 &= T_{force} / (S_{tba} \cdot N_{bt}) \\
 &= 6197.31/(137.3 \cdot 2.0) \\
 &= 22.5704 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Required Area of a Single Bolt [Bltarear]:} \\
 &= \max[B_{ltarearl}, B_{ltarears}, B_{ltareart}] \\
 &= \max[0.0, 583.9539, 22.5704] \\
 &= 583.9539 \text{ mm}^2
 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned}
 \text{Bearing Pressure (fc)} \\
 &= 2(QO + T_{force}) / (Y \cdot B_{pwid}) \\
 &= 2(339344.12 + 6197.31)/(2230.05 \cdot 350.0) \\
 &= 0.89 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance from Baseplate Edge to the Web [ADIST]:} \\
 &= (B_{plen} - W_{blength}) / 2 \\
 &= (3400.0 - 3349.2)/2 \\
 &= 25.4000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Overturning Moment due To Bolt Tension [Mt]:} \\
 &= T_{force} \cdot A_{dist} \\
 &= 6197.31 \cdot 25.4 \\
 &= 157475.53 \text{ N-mm}
 \end{aligned}$$

Equivalent Bearing Pressure (f1):
 $= f_c \cdot (Y - Adist) / Y$
 $= 0.89 \cdot (2230.05 - 25.4) / 2230.05$
 $= 0.88 \text{ MPa}$

Overturning Moment due to Bearing Pressure [Mc]:
 $= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot f_c)$
 $= (25.4^2 \cdot 350.0 / 6) \cdot (0.88 + 2 \cdot 0.89)$
 $= 99626.95 \text{ N-mm}$

Baseplate Required Thickness [Treq]:
 $= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
 $= \sqrt{(6 \cdot \max(157475.53, 99626.95) / (350.0 \cdot 207.0))}$
 $= 3.6108 \text{ mm}$

MDMT Calculations for: 22, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.939$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-14 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

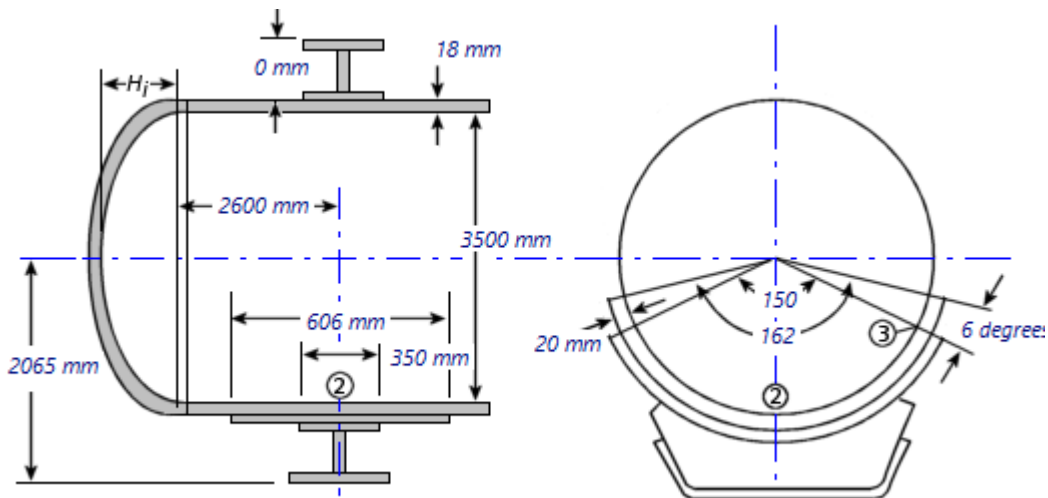
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1760.50	mm
Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	13000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2600.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	606.00	mm
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	885.00	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2065.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	909916.31	N
Pressure used in Analysis	P	1.442 MPa

Horizontal Vessel Analysis Results:		Actual MPa	Allowable MPa
Long. Stress at Top of Midspan		81.74	234.00
Long. Stress at Bottom of Midspan		87.49	234.00
Long. Stress at Top of Saddles		88.42	234.00
Long. Stress at Bottom of Saddles		80.81	234.00
Tangential Shear in Shell		6.03	138.58
Circ. Compressive Stress in Shell		6.73	234.00
Stiffener Circ. Stress at Shell		31.82	234.00
Stiffener Circ. Stress at Tip		172.91	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(36259.0/2 + 0) \cdot 2065.0/3400.0002$
 $= 33033.1 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(11479.28, 0) \cdot 2065.0/7800.0$
 $= 3039.1 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 876883 + \max(3039, 33033, 0, 0)$
 $= 909916.3 \text{ N}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1654.611(4.243 + \pi/4(3.626 \cdot 1.2)^2)$
 $= 34787.480 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	920607.50	N
Transverse Shear Load	18129.52	N
Longitudinal Shear Load	11479.28	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 3400.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	K6p = 0.0252
K7p = 0.0252			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -909916 \cdot 2600.0 [1 - (1 - 2600.0/13000.0 + (1760.5^2 - 885.0^2)/(2 \cdot 2600.0 \cdot 13000.0))/(1 + (4 \cdot 885.0)/(3 \cdot 13000.0))]$
 $= -556566400.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 909916 \cdot 13000/4((1 + 2(1760^2 - 885^2)/(13000^2))/(1 + (4 \cdot 885)/(3 \cdot 13000)) - 4 \cdot 2600/13000)$
 $= 419839808.0 \text{ N-mm}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 1.442 \cdot 1760.5/(2 \cdot 15.0) - 0.41984E+09/(\pi \cdot 1760.5^2 \cdot 15.0)$
 $= 81.74 \text{ MPa}$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 1.442 \cdot 1760.5/(2 \cdot 15.0) + 0.41984E+09/(\pi \cdot 1760.5^2 \cdot 15.0)$
 $= 87.49 \text{ MPa}$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:
 $= P \cdot R_m/(2t) - M_1/(\pi \cdot R_m^2 \cdot t)$
 $= 1.442 \cdot 1760.5/(2 \cdot 15.0) - 0.6E+09/(\pi \cdot 1760.5^2 \cdot 15.0)$
 $= 88.42 \text{ MPa}$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 t)$$

$$= 1.442 \cdot 1760.5 / (2 \cdot 15.0) + -0.6E+09 / (\pi \cdot 1760.5^2 \cdot 15.0)$$

$$= 80.81 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 909916(13000.0 - 2 \cdot 2600.0) / (13000.0 + (4 \cdot 885.0 / 3))$$

$$= 500518.2 \text{ N}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 500518 / (\pi \cdot 1760.4999 \cdot 15.0)$$

$$= 6.03 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1760.5 \cdot 15.0}$$

$$= 126.753 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.673 \cdot 909916 \cdot 0.1/9102.59$$

$$= -6.73 \text{ MPa}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.3021 \cdot 909916 / 9102.59 + 0.0317 \cdot 909916 \cdot 1760.5 \cdot 80.28 / 65682084$$

$$= 31.82 \text{ MPa}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.3021 \cdot 909916 / 9102.59 - 0.0317 \cdot 909916 \cdot 1760.5 \cdot 184.72 / 65682084$$

$$= -172.91 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3400.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	259.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$

$$= ((1.56 \cdot \sqrt{1753.0 \cdot 15.0}) + 606.0)^3 / 12$$

$$= 241583.797 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 241583.797 + 12884.48(88.832 - 7.5)^2$$

$$= 85470048.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	859.0	15.0	7.5	12884.5	96633.6	0.855E+08
Wearplate	606.0	20.0	25.0	12120.0	303000.0	0.498E+08
Web	10.0	255.0	162.5	2550.0	414375.4	0.277E+08
BasePlate	350.0	22.0	301.0	7700.0	2317701.2	0.347E+09
Totals	...	...	...	35254.5	3131710.2	0.510E+09



Distance to Centroid [C1]:

$$= AY / A$$

$$= 3131710/35254.48$$

$$= 88.832 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 150.0/2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 909916$$

$$= 236007.1250 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 10.5511 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2065.0 - 1753.0 \cdot \sin(150.0/2) / 1.309$$

$$= 771.439 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 182138864.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 31.7244 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt[3]{(3(909916 + 10691) \cdot 350.0 / (4 \cdot 3400.0 \cdot 173.33))}$$

$$= 20.251 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (1750.0 + 18.0 + 20.0)$$

$$= 3454.151 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 3454.1506 / (5 - 1)$$

$$= 863.538 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 863.5377 \cdot 350.0/2$$

$$= 151119.094 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 920608 / (3400.0 \cdot 350.0)$$

$$= 0.774 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 151119.1 \cdot 0.77$$

$$= 116908.711 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 3400.0 + 4317.688$$

$$= 7717.688 \text{ mm}^2$$

Compressive Stress [Sc]:

= P/A_r
 = 116908.7/7717.688
 = 15.149 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	3500.0	...	0.357E+08

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 350.0 - 10.0
 = 340.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/7717.688
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 350.0/2
 = 175.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{35729128/7717.688}$
 = 68.041 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - $\cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 = 2065.0 - $\cos(150.0/2) (1750.0 + 18.0 + 20.0) - 22.0$
 = 1580.232 mm

Length of Outer Rib, step 2, [Lw]:

= $2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 = $2 \cdot \sin(150.0/2) (1750.0 + 18.0 + 20.0)$
 = 3454.151 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 3400.0 - 50.8
 = 3349.200 mm

Length of Outer Rib, step 4, [Dd]:

= $(L_w - B_d) / 2$
 = $(3454.151 - 3349.2) / 2$
 = 52.475 mm

Length of Outer Rib [L]:

= $\sqrt{L_v^2 + D_d^2}$
 = $\sqrt{1580.232^2 + 52.475^2}$
 = 1581.103 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199955/260.0}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = $1 \cdot 1581.103/68.041$
 = 23.238

Bending Moment [Rm]:

= $Fl / (2 \cdot B_{plen}) \cdot e \cdot L / 2$

$$= 11479.3 / (2 \cdot 3400.0) \cdot 863.538 \cdot 1581.1/2$$

$$= 1152903.625 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (23.2377 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (23.24)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (23.24) / (8 \cdot 123.21) - (23.24)^3 / (8 \cdot 123.21^3))$$

$$= 147.1 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 15.15 / 147.06 + (1152904 \cdot 175.0 / 35729164) / 173.33$$

$$= 0.136$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	3400.0	0.0	0.357E+08
Web	863.5	10.0	0.0	8635.4	0.0	0.720E+05
Totals	...	...	...	12035.4	...	0.358E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 12035.376$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350.0 / 2$$

$$= 175.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 2065.0 - \cos(1788.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 255.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{35800256 / 12035.376}$$

$$= 54.540 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 255.0 / 54.54$$

$$= 4.675$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 11479.275 / (2 \cdot 3400.0)$$

$$= 1.688 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.688 \cdot 863.538 \cdot 255.0$$

$$= 371880.406 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 371880.438 / 204573.094$$

$$= 1.817 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (4.6755 < 123.2107) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.68)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (4.68) / (8 \cdot 123.21) - (4.68)^3 / (8 \cdot 123.21^3))$$

$$= 154.6 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 19.85/154.57 + (371880.44 \cdot 175.0/35800256)/173.33$$

$$= 0.139$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	738.0151 mm ²
Saddle Load QO (Weight)	QO	887574.4 N
Saddle Load QL (Wind/Seismic contribution)	QL	3039.1 N
Maximum Transverse Force	Ft	18129.5 N
Maximum Longitudinal Force	Fl	11479.3 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 11479.3 / (738.015 \cdot 4)$$

$$= 3.9 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 18129.5 / (738.015 \cdot 4)$$

$$= 6.1 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 11479.28 / (103.2 \cdot 4.0)$$

$$= 27.8112 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2065.0 \cdot 18129.52 + 0.0$$

$$= 37452648.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 37452648/887574$$

$$= 42.18 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 27.8112, 0.0]$$

$$= 27.8112 \text{ mm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Input and Calculated Values:

Vessel Mean Radius	Rm	1760.50 mm
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Shell Thickness used in this Case	t	18.000	mm
Stiffened Vessel Length per 4.15.6	L	13000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	10400.00	mm
Saddle Width	b1 or b	350.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Wear Plate Width	b2 or b1	606.00	mm
Wear Plate Bearing Angle	delta2 or theta1	162.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	MPa
Inside or Mean Depth of Head	Hi or hm	885.50	mm
Shell Allowable Stress used in Calculation		234.00	MPa
Head Allowable Stress used in Calculation		234.00	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2065.00	mm
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		903405.38	N
Pressure used in Analysis	P	1.442	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	130.01	234.00
Long. Stress at Bottom of Midspan	39.22	234.00
Long. Stress at Top of Saddles	136.65	234.00
Long. Stress at Bottom of Saddles	32.58	234.00
Tangential Shear in Shell	5.99	138.58
Circ. Compressive Stress in Shell	8.10	234.00
Stiffener Circ. Stress at Shell	31.89	234.00
Stiffener Circ. Stress at Tip	229.26	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (36259.0/2 + 0) \cdot 2065.0/3400.0002$
 $= 33033.1 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(11479.28, 0) \cdot 2065.0/7800.0$
 $= 3039.1 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 870372 + \max(3039, 33033, 0, 0)$
 $= 903405.4 \text{ N}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1654.611 (4.243 + \pi/4 (3.626 \cdot 1.2)^2)$
 $= 34787.480 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	914096.56	N
Transverse Shear Load	18129.52	N
Longitudinal Shear Load	11479.28	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length
 = 3400.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1607 K2 = 0.7988 K3 = 0.4851 K4 = 0.2952
 K5 = 0.6733 K6 = 0.0317 K7 = 0.0317 K8 = 0.3021
 K9 = 0.2177 K10 = 0.0355 K1* = 0.2792 K6p = 0.0252
 K7p = 0.0252

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 = $-903405 \cdot 10400.0 [1 - (1 - 10400.0/13000.0 + (1760.5^2 - 885.5^2)/(2 \cdot 10400.0 \cdot 13000.0))/(1 + (4 \cdot 885.5)/(3 \cdot 13000.0))]$
 = -7602114560.0 N-mm

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$
 = $903405 \cdot 13000/4 [(1 + 2(1760^2 - 886^2)/(13000^2))/(1 + (4 \cdot 886)/(3 \cdot 13000)) - 4 \cdot 10400/13000]$
 = -6632741376.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$
 = $1.442 \cdot 1760.5/(2 \cdot 15.0) - -0.7E+10/(\pi \cdot 1760.5^2 \cdot 15.0)$
 = 130.01 MPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $1.442 \cdot 1760.5/(2 \cdot 15.0) + -0.7E+10/(\pi \cdot 1760.5^2 \cdot 15.0)$
 = 39.22 MPa

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

= $P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t)$
 = $1.442 \cdot 1760.5/(2 \cdot 15.0) - -0.8E+10/(\pi \cdot 1760.5^2 \cdot 15.0)$
 = 136.65 MPa

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

= $P \cdot Rm/(2t) + M1/(\pi \cdot Rm^2 \cdot t)$
 = $1.442 \cdot 1760.5/(2 \cdot 15.0) + -0.8E+10/(\pi \cdot 1760.5^2 \cdot 15.0)$
 = 32.58 MPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot hm/3))$
 = $903405(13000.0 - 2 \cdot 10400.0)/(13000.0 + (4 \cdot 885.5/3))$
 = -496913.2 N

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

= $T / (\pi \cdot Rm \cdot t)$
 = $-496913.25/(\pi \cdot 1760.4999 \cdot 15.0)$
 = -5.99 MPa

Decay Length (4.15.20) [x_1, x_2]:

= $0.78 \cdot \sqrt{(Rm \cdot t)}$
 = $0.78 \cdot \sqrt{(1760.5 \cdot 15.0)}$
 = 126.753 mm

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_{6*}]:

= $-K5 \cdot Q \cdot k/A$
 = $-0.673 \cdot 903405 \cdot 0.1/7512.59$
 = -8.10 MPa

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_{8*}]:

= $-K8 \cdot Q/A + K6 \cdot Q \cdot Rm \cdot c_1/l$



$$= -0.3021 \cdot 903405 / 7512.59 + 0.0317 \cdot 903405 \cdot 1760.5 \cdot 69.23 / 51124100$$

$$= 31.89 \text{ MPa}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.3021 \cdot 903405 / 7512.59 - 0.0317 \cdot 903405 \cdot 1760.5 \cdot 195.77 / 51124100$$

$$= -229.26 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	3400.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	350.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	MPa
Height of Web at Center	Hw,c	259.0	mm
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) \cdot t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1753.0 \cdot 15.0)} + 606.0) \cdot 15.0^3) / 12$$

$$= 241583.797 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 241583.797 + 12884.48 (88.832 - 7.5)^2$$

$$= 85470048.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	859.0	15.0	7.5	12884.5	96633.6	0.855E+08
Wearplate	606.0	20.0	25.0	12120.0	303000.0	0.498E+08
Web	10.0	255.0	162.5	2550.0	414375.4	0.277E+08
BasePlate	350.0	22.0	301.0	7700.0	2317701.2	0.347E+09
Totals	...	...	...	35254.5	3131710.2	0.510E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3131710 / 35254.48$$

$$= 88.832 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 150.0/2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.259 \cdot 903405$$

$$= 234318.3750 \text{ N}$$

Tension Stress, $St = (Fh / A_s) = 10.4756 \text{ MPa}$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ MPa}$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2065.0 - 1753.0 \cdot \sin(150.0/2) / 1.309$$

$$= 771.439 \text{ mm}$$

Bending Moment, $M = F_h \cdot d = 180835568.0000 \text{ N-mm}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 31.4974 \text{ MPa}$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ MPa}$

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt[3]{(3(903405 + 10691)350.0 / (4 \cdot 3400.0 \cdot 173.33))}$
 $= 20.179 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 150.0/2) (1750.0 + 18.0 + 20.0)$
 $= 3454.151 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 3454.1506 / (5 - 1)$
 $= 863.538 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 863.5377 \cdot 350.0 / 2$
 $= 151119.094 \text{ mm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 914097 / (3400.0 \cdot 350.0)$
 $= 0.768 \text{ N/mm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 151119.1 \cdot 0.77$
 $= 116081.898 \text{ N}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 3400.0 + 4317.688$
 $= 7717.688 \text{ mm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 116081.9 / 7717.688$
 $= 15.042 \text{ MPa}$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	350.0	...	3500.0	...	0.357E+08

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 350.0 - 10.0$
 $= 340.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 7717.688$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 350.0 / 2$
 $= 175.000 \text{ mm}$



Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{35729128 / 7717.688}$
 $= 68.041 \text{ mm}$

Length of Outer Rib, step 1, [Lv]:
 $= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 2065.0 - \cos(150.0/2) (1750.0 + 18.0 + 20.0) - 22.0$
 $= 1580.232 \text{ mm}$

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(150.0/2) (1750.0 + 18.0 + 20.0)$
 $= 3454.151 \text{ mm}$

Length of Outer Rib, step 3, [Bd]:
 $= \text{Baseplate length} - \text{clearance}$
 $= 3400.0 - 50.8$
 $= 3349.200 \text{ mm}$

Length of Outer Rib, step 4, [Dd]:
 $= (\text{Lw} - \text{Bd}) / 2$
 $= (3454.151 - 3349.2) / 2$
 $= 52.475 \text{ mm}$

Length of Outer Rib [L]:
 $= \sqrt{\text{Lv}^2 + \text{Dd}^2}$
 $= \sqrt{1580.232^2 + 52.475^2}$
 $= 1581.103 \text{ mm}$

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199955 / 260.0}$
 $= 123.211$

Slenderness ratio [KL/r]:
 $= \text{KL} / \text{r}$
 $= 1 \cdot 1581.103 / 68.041$
 $= 23.238$

Bending Moment [Rm]:
 $= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 $= 11479.3 / (2 \cdot 3400.0) \cdot 863.538 \cdot 1581.1 / 2$
 $= 1152903.625 \text{ N-mm}$

Compressive Allowable, $\text{KL}/\text{r} < \text{Cc}$ ($23.2377 < 123.2107$) per AISC E2-1 [Sca]:
 $= (1 - (\text{KL}/\text{r})^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/\text{r}) / (8 \cdot \text{Cc}) - (\text{KL}/\text{r})^3 / (8 \cdot \text{Cc}^3))$
 $= (1 - (23.24)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (23.24) / (8 \cdot 123.21) - (23.24^3) / (8 \cdot 123.21^3))$
 $= 147.1 \text{ MPa}$

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= \text{Sc} / \text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$
 $= 15.04 / 147.06 + (1152904 \cdot 175.0 / 35729164) / 173.33$
 $= 0.135$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	340.0	0.0	3400.0	0.0	0.357E+08
Web	863.5	10.0	0.0	8635.4	0.0	0.720E+05
Totals	...	...	...	12035.4	...	0.358E+08

Distance to Centroid from Datum [ytot]:
 $= \text{AY} / \text{A}$
 $= 0.0 / 12035.376$
 $= 0.000 \text{ mm}$



Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 350.0 / 2$$

$$= 175.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 2065.0 - \cos(1788.0/2)(22.0 + 0.0 + 0.0) - 0.0$$

$$= 255.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(35800256 / 12035.376)}$$

$$= 54.540 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 255.0 / 54.54$$

$$= 4.675$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 11479.275 / (2 \cdot 3400.0)$$

$$= 1.688 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.688 \cdot 863.538 \cdot 255.0$$

$$= 371880.406 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 371880.438 / 204573.094$$

$$= 1.817 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($4.6755 < 123.2107$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.68)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (4.68) / (8 \cdot 123.21) - (4.68)^3 / (8 \cdot 123.21^3))$$

$$= 154.6 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 19.71 / 154.57 + (371880.44 \cdot 175.0 / 35800256) / 173.33$$

$$= 0.138$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 MPa
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	36.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 MPa
Area Available in a Single Bolt	BltArea	738.0151 mm ²
Saddle Load Q0 (Weight)	Q0	881063.5 N
Saddle Load QL (Wind/Seismic contribution)	QL	3039.1 N
Maximum Transverse Force	Ft	18129.5 N
Maximum Longitudinal Force	Fl	11479.3 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 11479.3 / (738.015 \cdot 4)$$

$$= 3.9 \text{ MPa. Must be less than } 103.2 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:



= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $18129.5 / (738.015 \cdot 4)$
= 6.1 MPa. Must be less than 103.2 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_t / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
= $11479.28 / (103.2 \cdot 4.0)$
= 27.8112 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $2065.0 \cdot 18129.52 + 0.0$
= 37452648.00 N-mm

Eccentricity (e):
= $Rmom / QO$
= $37452648 / 881064$
= 42.49 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 27.8112, 0.0]$
= 27.8112 mm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N1 40"

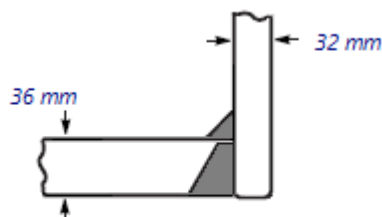
From: 20

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	36.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1160.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		40.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	32.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N1 40"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	40.000 in.
--	------------

Actual Thickness Used in Calculation 1.260 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 511.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 4.0773 \text{ mm}$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 2044.0000 mm
 Parallel to Vessel Wall, opening length d 1022.0000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 72.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

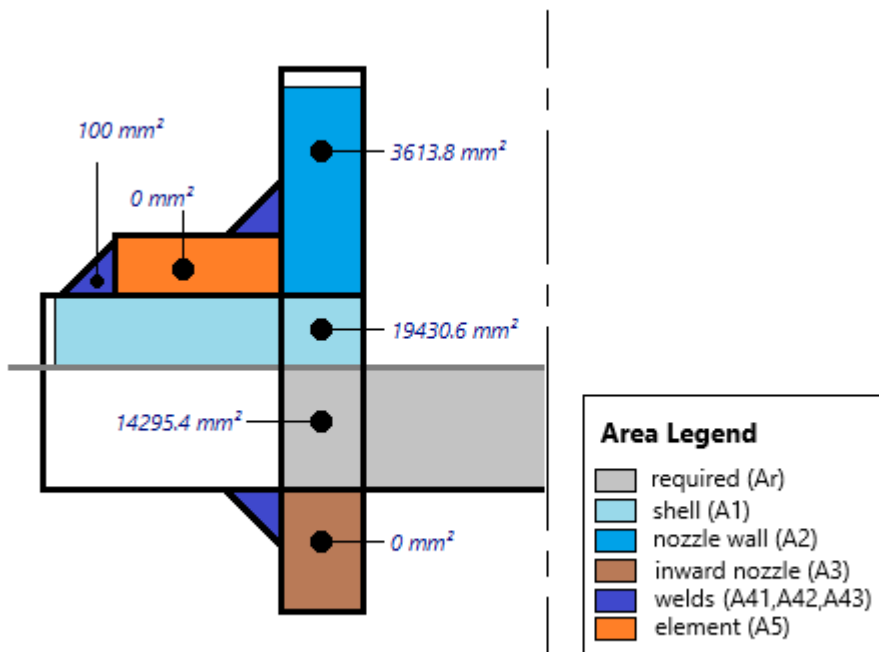


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	14295.440	NA	NA
Area in Shell	A1	19430.561	NA	NA
Area in Nozzle Wall	A2	3613.784	NA	NA



Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	23144.344	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (1022.0 \cdot 13.9877 \cdot 1.0 + 2 \cdot 29.0 \cdot 13.9877 \cdot 1.0 \cdot (1 - 1.0))$
 $= 14295.440 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 1022.0(1.0 \cdot 33.0 - 1.0 \cdot 13.988) - 2 \cdot 29.0$
 $(1.0 \cdot 33.0 - 1.0 \cdot 13.9877) \cdot (1 - 1.0)$
 $= 19430.561 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 72.5)(29.0 - 4.08)1.0$
 $= 3613.784 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 7.0773 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 16.9877 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 16.9877 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.331, \max(16.9877, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(7.0773, 11.3312)$
 $= 11.3312 \text{ mm}$

Available Nozzle Neck Thickness = 32.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17.3,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 327.7 MPa	Passed
Occasional	: 9.4,	Allowable	: 183.5 MPa	Passed
Shear	: 5.4,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 32.0, tr = 4.077, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.141$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $1.10/5.10 = 0.215$

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1533.0000 mm
 Parallel to Vessel Wall, opening length d 766.5000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 72.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	9530.771	9530.771
Area Available in Shell	A1	9715.280	9715.280
Area Available in Nozzle Wall	A2	3613.784	3613.784
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	100.000	100.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	13429.064	13429.064

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

*Warning: Flange Dimensions not found ...
 The flange could not be considered in the moment of inertia considerations.*

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad ²
Nozzle	9657.0	166.5	0.8924E+08	96.510	0.1792E+09
Shell	17424.0	16.5	0.1581E+07	53.490	0.5143E+08
Sum	27081.0				0.2306E+09

Membrane stress per Appendix 1-7 (1 or 2) Sm 127.02 MPa
 Allowable Membrane Stress S 138.00 MPa
 Bending stress at the inside of the Nozzle Sb 23.33 MPa
 Combined Membrane plus Bending Stress Sm + Sb 150.35 MPa
 Allowable Membrane + Bending Stress 1.5 * S 207.00 MPa
 M.A.W.P for the Appendix 1-7 Stress calc 1.19 MPa

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 4537.002 mm²
 Area from Shell = 7974.367 mm²
 Total Area, As = 12511.369 mm²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P[R(Rn + tn + \sqrt{(rm \cdot t)}) + Rn(t + \sqrt{(Rnm \cdot tn)})] / As$$

$$= 1.096[1753.0(511.0 + 29.0 + 241.647) +$$

$$511.0(156.448) / 12511.369$$

$$= 127.021 \text{ MPa}$$

The area A_s was computed per figure 1-7-1 for the S_m calculation.

Moment [M]:

$$= (R_n^3/6 + R \cdot R_n \cdot e)P$$

$$= (511.0^3/6 + 1753.0 \cdot 511.0 \cdot 53.49)1.096$$

$$= 76904560 \text{ N-mm}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= (M(e + t/2)) / I$$

$$= (76904560(53.49 + 33.0/2)) / .23062E+09$$

$$= 23.332 \text{ MPa}$$

Weld Size Calculations, Description: N1 40"

Intermediate Calc. for nozzle/shell welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r))S_v)$$

$$= \max(0, (14295.4404 - 19430.5605 + 2 \cdot 29.0 \cdot 1.0 \cdot (1.0 \cdot 33.0 - 13.9877))138)$$

$$= \max(0, -556414.94) \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - C_{an}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (3613.7839 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 512449.34 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (3613.7839 + 0.0 + 100.0 + (1914.0)) \cdot 138$$

$$= 776554.06 \text{ N}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (3613.7839 + 0.0 + 100.0 + 0.0 + (1914.0)) \cdot 138$$

$$= 776554.06 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 1080.0 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 1147028. \text{ N}$$

Shear, Nozzle Wall [S_{nw}]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 525.5) \cdot (32.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 4624457. \text{ N}$$

Tension, Shell Groove Weld [T_{ngw}]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 1080.0 \cdot (24.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 4157466. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:



$$= (\text{SONW} + \text{SNW}) = (1147028 + 4624457) = 5771485 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (1147028 + 0 + 4157466 + 0) = 5304494 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (1147028 + 4157466 + 0) = 5304494 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 5771484 N , must exceed W = 0 N	or W1 = 512449 N
Path 2-2 = 5304494 N , must exceed W = 0 N	or W2 = 776554 N
Path 3-3 = 5304494 N , must exceed W = 0 N	or W3 = 776554 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 85.3979 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 421.3978 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 3.053 %

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N1 40" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3500.000	mm
Shell Thickness	36.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	1016.000	mm
Nozzle Thickness	32.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	MPa
Nozzle Hot S.I. Allowable	SNmh 138.00	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1.10	MPa
Radial Load	P	-26999.98	N
Circumferential Shear	Vc	40000.02	N
Longitudinal Shear	Vl	40000.02	N
Circumferential Moment	Mc	0.120E+09	N-mm
Longitudinal Moment	Ml	0.160E+09	N-mm
Torsional Moment	Mt	0.200E+09	N-mm

Include Axial Pressure Thrust No



Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 1080.0 + 2 \cdot 1.65 \cdot \sqrt{(1769.5(36.0 - 3.0))}$
 $= 1877.437 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3539.000 mm
 Nozzle Outside Diameter (d) = 1080.000 mm
 Vessel Thickness used (T) = 33.000 mm
 Nozzle Thickness used (t) = 29.000 mm
 T / t = 1.138
 d / t = 37.241

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.160$$

Nr/P = 0.043
 Mr/P = 0.093
 M0/P = 0.033
 N0/P = 0.128
 MrD/Mc = 0.221
 NrDL/Mc = 0.076
 M0d/Mc = 0.084
 N0DL/Mc = 0.138
 MrD/Ml = 0.084
 NrDL/Ml = 0.037
 M0D/Ml = 0.030
 N0DL/Ml = 0.165

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-22	-22	22	22
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	57	58	57	58
Outplane Stress Summary	18	58	112	55

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-68	68	68	-68
Inplane Pressure Stress	28	28	28	28
Inplane Stress Summary	-30	79	117	-47

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	49	80	118	103

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-51	51	51	-51
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	28	28	28	28
Outplane Stress Summary	-28	64	102	-10

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-135	135	135	-135
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	57	58	57	58
Inplane Stress Summary	-70	173	215	-81

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	70	173	215	81

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0



Hoop Membrane (ML)	-22	-22	22	22
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	23	22	23
Hoop Stress Summary	3	4	48	49

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-69	69	69	-69
Axial Pressure Stress	11	11	11	11
Axial Stress Summary	-50	60	101	-65

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	54	61	101	114

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-14	-14	14	14
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	23	22	23
Hoop Stress Summary	11	12	39	41

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	13	-13	13	-13
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-144	144	144	-144
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11
Axial Stress Summary	-124	137	175	-142

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3

Shear Stress Summary	2	2	4	4
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	136	137	175	183

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	57	58	57	58	57	58	57	58
Circ. Pl (SUS)	-19	-19	25	25	-6	-6	8	8
Circ. Q (SUS)	-20	20	28	-28	-122	122	148	-148
Long. Pm (SUS)	28	28	28	28	28	28	28	28
Long. Pl (SUS)	-3	-3	5	5	-11	-11	17	17
Long. Q (SUS)	-55	55	81	-81	-47	47	55	-55
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	57.0	58.0	57.0	58.0	57.0	58.0	57.0	58.0
Pm+Pl (SUS)	38.0	39.0	82.0	83.0	51.0	52.0	65.0	66.0
Pm+Pl+Q (Total)	48.4	80.4	115.6	103.2	71.2	174.1	213.1	82.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.00	138.00	Passed
Pm+Pl (SUS)	83.00	207.00	Passed
Pm+Pl+Q (TOTAL)	213.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	23	22	23	22	23	22	23
Circ. Pl (SUS)	-19	-19	25	25	-11	-11	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-6	-6	6	6	-4	-4	4	4
Long. Q (SUS)	-56	56	82	-82	-131	131	157	-157
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3



Pm (SUS)	22.0	23.0	22.0	23.0	22.0	23.0	22.0	23.0
Pm+Pl (SUS)	5.0	5.0	47.0	48.0	11.0	12.0	39.0	40.0
Pm+Pl+Q (Total)	54.3	61.2	99.2	113.2	135.1	138.1	172.1	182.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23.00	138.00	Passed
Pm+Pl (SUS)	48.00	207.00	Passed
Pm+Pl+Q (TOTAL)	182.10	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N4 1"

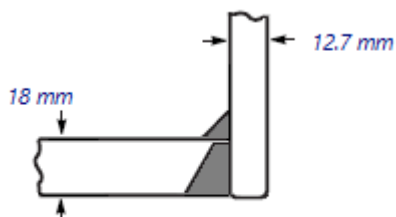
From: 20

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2300.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N4 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.000	in.
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Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([1.1 / (138 \cdot 1.0)] - 1))$
 $= 0.1252 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	80.8000 mm
Parallel to Vessel Wall	Rn+tn+t	40.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.2500 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.1252 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 16.9877 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 16.9877 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(16.9877, 4.5)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.1252, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 12.7000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 12.7, tr = 0.125, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.013$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N4 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.7000 mm
---	------	-----------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 405-V-003_7% Page 104 of 186
Nozzle Calcs.: N4 1" Nozl: 34 2:37pm Jul 20,2026



HEXAGON

The Drop for this Nozzle is : 0.1843 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.1844 mm

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Input, Nozzle Desc: N5A 1"

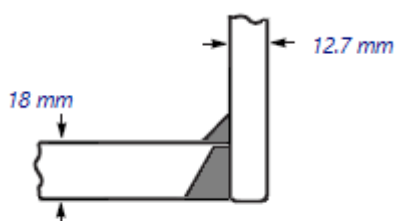
From: 20

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	500.0000	mm
Distance from Bottom/Left Tangent		1000.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-16.43	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: N5A 1"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.000 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([1.1 / (138.0 \cdot 1.0)] - 1))$
 $= 0.1252 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	82.1499	mm
Parallel to Vessel Wall	Rn+tn+t	41.0750	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N5A 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, tg = 12.7, tr = 0.125, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.013$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N5A 1"

Intermediate Calc. for nozzle/shell Welds Tmin 9.7000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: N5A 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.000 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]



= $R \left(\exp\left[\frac{P}{S_n \cdot E}\right] - 1 \right)$ per Appendix 1-2 (a)(1)
 = $15.7 \left(\exp\left[\frac{1.1}{138.0 \cdot 1.0}\right] - 1 \right)$
 = 0.1252 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	80.8000	mm
Parallel to Vessel Wall	Rn+tn+t	40.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N5A 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.1252	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 16.9877	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 16.9877	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[6.42, max(16.9877, 4.5)]
 = 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.1252, 6.42)
 = 6.4200 mm

Available Nozzle Neck Thickness = 12.7000 mm -> OK

Weld Size Calculations, Description: N5A 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.7000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 7.7832 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 326.5579 mm

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Input, Nozzle Desc: N5B 1"

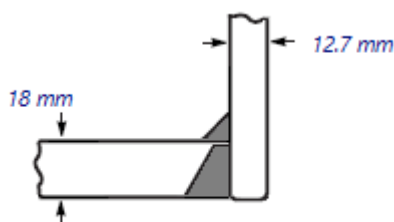
From: 20

Pressure for Reinforcement Calculations	P	1.1031	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1000.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N5B 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.000	in.
Actual Thickness Used in Calculation	0.500	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 14.0800 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([1.1 / (138.0 \cdot 1.0)] - 1))$
 $= 0.1260 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	80.8000	mm
Parallel to Vessel Wall	Rn+tn+tt	40.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N5B 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.1260	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.0800	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.0800	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(17.08, 4.5)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.126, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 12.7000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 12.7, tr = 0.126, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.013$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N5B 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.7000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1843 mm

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 405-V-003_7% Page 110 of 186
Nozzle Calcs.: N5B 1" Nozl: 36 2:37pm Jul 20,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.1844 mm

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Input, Nozzle Desc: N9 8"

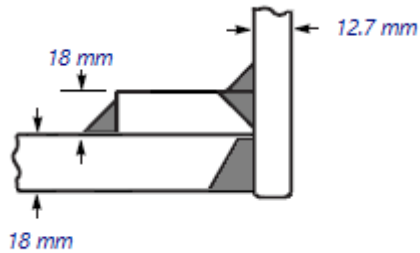
From: 20

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2800.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	359.0750	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N9 8"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 99.84 / (118 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 0.9324 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500 mm
Parallel to Vessel Wall, opening length	d	199.6750 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

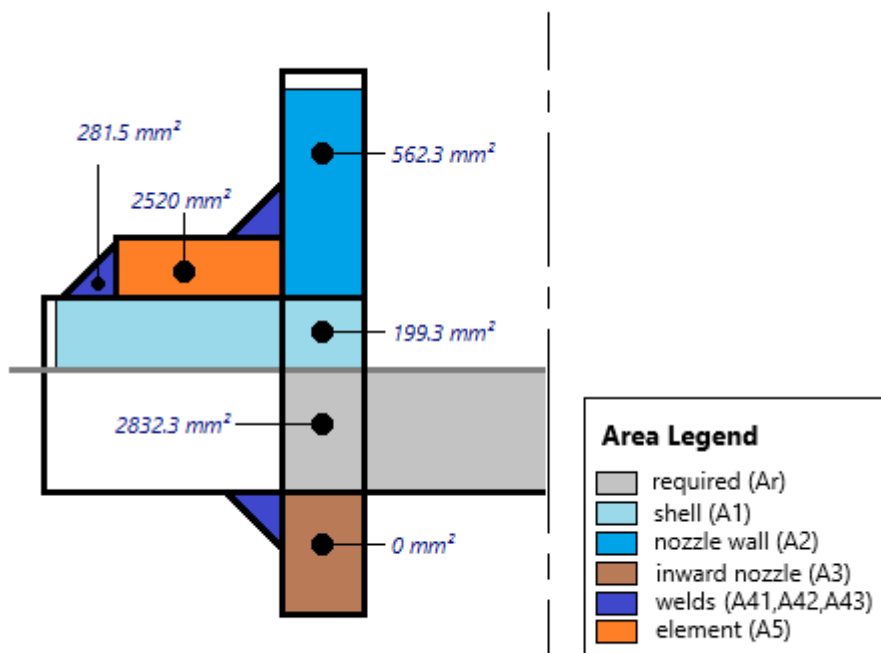


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2832.320	NA	NA
Area in Shell	A1	199.283	NA	NA
Area in Nozzle Wall	A2	562.280	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	281.509	NA	NA
Area in Element	A5	2520.000	NA	NA
TOTAL AREA AVAILABLE	Atot	3563.072	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 318.4776 18.0000 mm
 Based on given Pad Diameter: 359.0750 12.7803 mm
 Based on Shell or Nozzle Thickness: 359.9606 12.7000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (199.675 \cdot 13.9877 \cdot 1.0 + 2 \cdot 9.7 \cdot 13.9877 \cdot 1.0 \cdot (1 - 0.86))$
 $= 2832.320 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 199.675(1.0 \cdot 15.0 - 1.0 \cdot 13.988) - 2 \cdot 9.7$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.9877) \cdot (1 - 0.855)$
 $= 199.283 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (9.7 - 0.93) \cdot 0.8551$

= 562.280 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 14.0^2 \cdot 1.0$
 = 281.509 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(359.075 - 219.075) \cdot 18.0 \cdot 1.0$
 = 2520.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9324 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 16.9877 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 16.9877 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[10.16, max(16.9877, 4.5)]
 = 10.1600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.9324, 10.16)
 = 10.1600 mm

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	29.5,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	265.5 MPa	Passed
Occasional	:	5.4,	Allowable	:	156.9 MPa	Passed
Shear	:	16.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.113, tr = 0.932, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.115$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18.0, tr = 13.988, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.933$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 11.113, tr = 0.932, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.115$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $1.10/5.10 = 0.215$

Weld Size Calculations, Description: N9 8"

Intermediate Calc. for nozzle/shell Welds Tmin 9.7000 mm
 Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	7.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2832.3196 - 199.2831 + 2 \cdot 9.7 \cdot 0.8551 \cdot (1.0 \cdot 15.0 - 13.9877)) \cdot 138)$
 $= 365638.72 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (562.2805 + 2520.0 + 281.5088 - 0.0 \cdot 0.86) \cdot 138$
 $= 464155.06 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (562.2805 + 0.0 + 85.5088 + (248.8306)) \cdot 138$
 $= 123720.78 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (562.2805 + 0.0 + 281.5088 + 2520.0 + (248.8306)) \cdot 138$
 $= 498490.12 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 198955. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 359.075 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 533904. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 104.6875) \cdot (12.7 - 3.0) \cdot 0.7 \cdot 118$$

$$= 263487. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 219.075 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 632498. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 527072. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (533904 + 263487) = 797392 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198955 + 632498 + 527072 + 0) = 1358525 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (533904 + 527072 + 0) = 1060976 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 797391 N	, must exceed W = 365638 N	or W1 = 464155 N
Path 2-2 = 1358524 N	, must exceed W = 365638 N	or W2 = 123720 N
Path 3-3 = 1060976 N	, must exceed W = 365638 N	or W3 = 498490 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.4316 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 271.4316 mm

Input Echo, WRC297 Item 1, Description: N9 8" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3500.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	193.675	mm



Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	359.075	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	1.10	MPa
Radial Load	P	-5400.01	N
Circumferential Shear	Vc	8000.00	N
Longitudinal Shear	Vl	8000.00	N
Circumferential Moment	Mc	0.480E+07	N-mm
Longitudinal Moment	Ml	0.640E+07	N-mm
Torsional Moment	Mt	0.800E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(1769.5(18.0 - 3.0))}$
 $= 756.707 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3539.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	33.000	mm
Nozzle Thickness used	(t) =	9.700	mm
T / t	=	3.402	
d / t	=	22.585	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.641$$

Nr/P	=	0.109
Mr/P	=	0.121
M0/P	=	0.131
N0/P	=	0.268
MrD/Mc	=	0.168
NrDL/Mc	=	0.134
M0d/Mc	=	0.225
N0DL/Mc	=	0.096
MrD/Ml	=	0.154
NrDL/Ml	=	0.127
M0D/Ml	=	0.200
N0DL/Ml	=	0.201

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5



Outplane Bending (ML)	-32	32	32	-32
Normal Pressure Stress	57	58	57	58

Outplane Stress Summary	25	82	100	29
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-24	24	24	-24
Inplane Pressure Stress	28	28	28	28
Inplane Stress Summary	4	47	61	4

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	3	3	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	26	83	100	29

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-27	27	27	-27
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	28	28	28	28
Outplane Stress Summary	4	51	63	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-20	20	20	-20
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	57	58	57	58
Inplane Stress Summary	38	73	84	38

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	3	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	39	73	85	38

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	14	12	14
Hoop Stress Summary	8	10	19	20

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	22	-22	22	-22
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-19	-19	19	19
Axial Bending (ML)	-169	169	169	-169
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-159	133	219	-164

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	8	8

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	170	135	219	186

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	14	12	14

Hoop Stress Summary	12	13	16	17
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	22	-22	22	-22
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-140	140	140	-140
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-125	109	185	-140
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10

Shear Stress Summary	8	8	13	13
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	138	110	186	160
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{1760.5(18.0 - 3.0)}$
 $= 755.338 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	3521.000 mm
Nozzle Outside Diameter (d)	=	359.075 mm
Vessel Thickness used (T)	=	15.000 mm
Nozzle Thickness used (t)	=	79.700 mm
T / t	=	0.188
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.562$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.084
Mr/P	=	0.173
M0/P	=	0.050
N0/P	=	0.089
MrD/Mc	=	0.299
NrDL/Mc	=	0.086
M0d/Mc	=	0.092
N0DL/Mc	=	0.066
MrD/M1	=	0.137
NrDL/M1	=	0.099
M0D/M1	=	0.042



NØDL/M1 = 0.107

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	127	128	127	128
Outplane Stress Summary	108	135	165	111

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	17	98	163	-16

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	108	135	166	128

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-32	32	32	-32
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63
Outplane Stress Summary	36	87	109	29

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-106	106	106	-106
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	127	128	127	128

Inplane Stress Summary	42	207	265	4
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		43	207	266	30

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	57	58	57	58	57	58	57	58
Circ. Pl (SUS)	-4	-4	6	6	-2	-2	2	2
Circ. Q (SUS)	-29	29	35	-35	-17	17	23	-23
Long. Pm (SUS)	28	28	28	28	28	28	28	28
Long. Pl (SUS)	-3	-3	3	3	0	0	2	2
Long. Q (SUS)	-21	21	27	-27	-24	24	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	57.0	58.0	57.0	58.0	57.0	58.0	57.0	58.0
Pm+Pl (SUS)	53.0	54.0	63.0	64.0	55.0	56.0	59.0	60.0
Pm+Pl+Q (Total)	24.4	83.2	98.2	29.4	38.3	73.4	82.4	37.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.00	138.00	Passed
Pm+Pl (SUS)	64.00	207.00	Passed
Pm+Pl+Q (TOTAL)	98.22	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	127	128	127	128	127	128	127	128
Circ. Pl (SUS)	-6	-6	10	10	-3	-3	7	7
Circ. Q (SUS)	-13	13	27	-27	-82	82	130	-130
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-5	-5	9	9	-1	-1	5	5
Long. Q (SUS)	-41	41	89	-89	-25	25	39	-39
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	127.0	128.0	127.0	128.0	127.0	128.0	127.0	128.0
Pm+Pl (SUS)	121.0	122.0	137.0	138.0	124.0	125.0	134.0	135.0
Pm+Pl+Q (Total)	108.0	135.1	165.0	128.1	42.7	207.0	264.0	29.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	128.00	138.00	Passed
Pm+Pl (SUS)	138.00	207.00	Passed
Pm+Pl+Q (TOTAL)	264.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	14	12	14	12	14	12	14
Circ. Pl (SUS)	-4	-4	6	6	0	0	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-19	-19	19	19	-14	-14	14	14
Long. Q (SUS)	-147	147	191	-191	-118	118	162	-162
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	12.0	14.0	12.0	14.0	12.0	14.0	12.0	14.0
Pm+Pl (SUS)	21.4	23.3	25.5	25.7	20.4	22.4	20.6	20.8
Pm+Pl+Q (Total)	169.7	135.2	216.3	186.7	138.9	110.7	182.9	159.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+Pl (SUS)	25.70	177.00	Passed
Pm+Pl+Q (TOTAL)	216.32	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



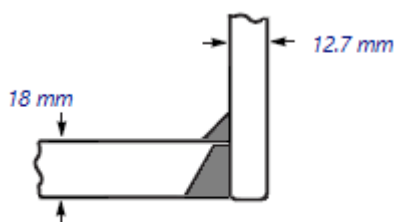
Input, Nozzle Desc: N11 1"
From: 20

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2450.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: N11 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.000	in.
Actual Thickness Used in Calculation	0.500	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([1.1 / (138.0 \cdot 1.0)] - 1))$
 $= 0.1252 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	80.8000	mm
Parallel to Vessel Wall	Rn+tn+tt	40.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N11 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.1252	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 16.9877	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 16.9877	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(16.9877, 4.5)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.1252, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 12.7000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.7, tr = 0.125, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.013$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N11 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.7000	mm
---	------	--------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1843 mm

PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 405-V-003_7% Page 127 of 186
Nozzle Calcs.: N11 1" Nozl: 38 2:37pm Jul 20,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.1844 mm

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Input, Nozzle Desc: N2 32"

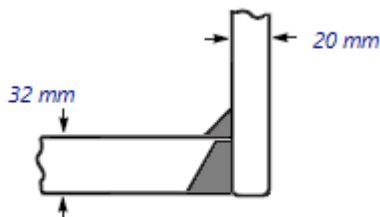
From: 50

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12040.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		32.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N2 32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	32.000	in.
--	--------	-----

Actual Thickness Used in Calculation 0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 409.4 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 3.2667 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1637.6000 mm
 Parallel to Vessel Wall, opening length d 818.8000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 42.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

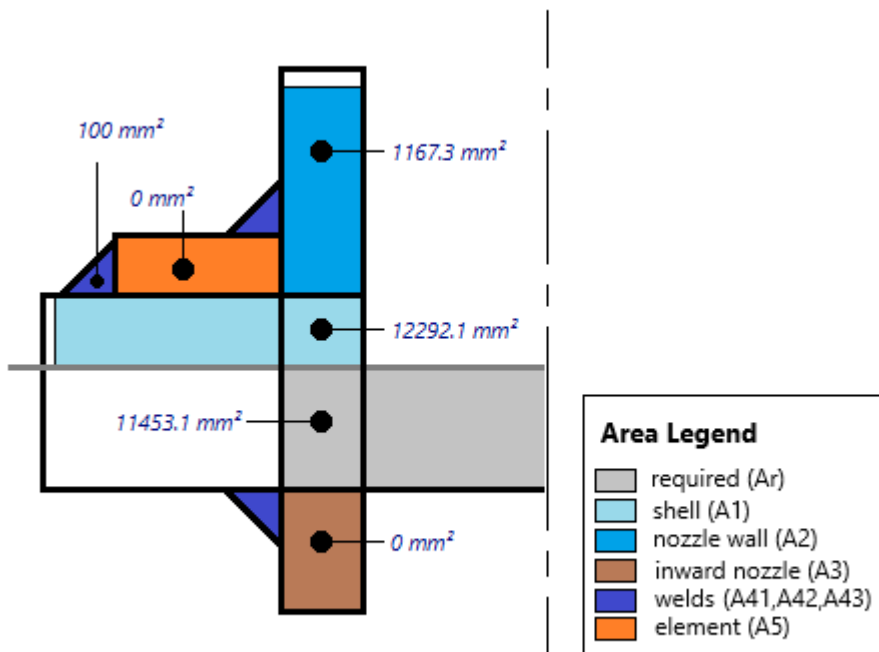


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	11453.138	NA	NA
Area in Shell	A1	12292.061	NA	NA
Area in Nozzle Wall	A2	1167.333	NA	NA



Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	13559.394	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (818.8 \cdot 13.9877 \cdot 1.0 + 2 \cdot 17.0 \cdot 13.9877 \cdot 1.0 \cdot (1 - 1.0))$
 $= 11453.138 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 818.8(1.0 \cdot 29.0 - 1.0 \cdot 13.988) - 2 \cdot 17.0$
 $(1.0 \cdot 29.0 - 1.0 \cdot 13.9877) \cdot (1 - 1.0)$
 $= 12292.061 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 42.5)(17.0 - 3.27)1.0$
 $= 1167.333 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 6.2667 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 16.9877 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 16.9877 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.331, \max(16.9877, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(6.2667, 11.3312)$
 $= 11.3312 \text{ mm}$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.4,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 318.6 MPa	Passed
Occasional	: 12.9,	Allowable	: 183.5 MPa	Passed
Shear	: 9.2,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 20.0, tr = 3.267, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.192$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $1.10/5.10 = 0.215$

Weld Size Calculations, Description: N2 32"

Intermediate Calc. for nozzle/shell Welds Tmin 17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (11453.1377 - 12292.0605 + 2 \cdot 17.0 \cdot 1.0 \cdot (1.0 \cdot 29.0 - 13.9877)) \cdot 138)$
 $= \max(0, -45329.02) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1167.3333 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 174873.97 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1167.3333 + 0.0 + 100.0 + (986.0)) \cdot 138$
 $= 310927.91 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1167.3333 + 0.0 + 100.0 + 0.0 + (986.0)) \cdot 138$
 $= 310927.91 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 852.8 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 905727. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 417.9) \cdot (20.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 2155814. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 852.8 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 138$

= 2462144. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 = (SONW + SNW) = (905727 + 2155814) = 3061541 N

Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (905727 + 0 + 2462144 + 0) = 3367871 N

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (905727 + 2462144 + 0) = 3367871 N

Summary of Failure Path Calculations:

Path 1-1 = 3061541 N , must exceed W = 0 N	or W1 = 174873 N
Path 2-2 = 3367871 N , must exceed W = 0 N	or W2 = 310927 N
Path 3-3 = 3367871 N , must exceed W = 0 N	or W3 = 310927 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 52.7425 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 584.7425 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 2.402 %

This nozzle or its diameter limit intersects weld seam # [4] on this element. The distance between the nozzle and seam is, 477.93 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 818.80 mm

Input Echo, WRC297 Item 1, Description: N2 32" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3500.000	mm
Shell Thickness	32.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	812.800	mm
Nozzle Thickness	20.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1.10	MPa
Radial Load	P	-21600.02	N



Circumferential Shear	Vc	31999.98	N
Longitudinal Shear	Vl	31999.98	N
Circumferential Moment	Mc	0.768E+08	N-mm
Longitudinal Moment	Ml	0.102E+09	N-mm
Torsional Moment	Mt	0.128E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 852.8 + 2 \cdot 1.65 \cdot \sqrt{(1767.5(32.0 - 3.0))}$
 $= 1599.924 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3535.000 mm
Nozzle Outside Diameter	(d) =	852.800 mm
Vessel Thickness used	(T) =	29.000 mm
Nozzle Thickness used	(t) =	17.000 mm
T / t	=	1.706
d / t	=	50.165

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.664$$

Nr/P	=	0.037
Mr/P	=	0.081
M0/P	=	0.037
N0/P	=	0.143
MrD/Mc	=	0.189
NrDL/Mc	=	0.078
M0d/Mc	=	0.099
N0DL/Mc	=	0.192
MrD/Ml	=	0.081
NrDL/Ml	=	0.039
M0D/Ml	=	0.036
N0DL/Ml	=	0.238

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-33	-33	33	33
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	65	66	65	66
Outplane Stress Summary	10	61	139	68

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-69	69	69	-69
Inplane Pressure Stress	32	32	32	32



Inplane Stress Summary -29 85 121 -42

Vessel Stresses (MPa):

LONGITUDINAL PLANE Au Al Bu Bl
 (Shear stress normal to Top Top Bottom Bottom
 longitudinal plane) Outside Inside Outside Inside

Outplane Shear (Vc) 0 0 0 0
 Outplane Shear (Vl) 0 0 0 0
 Outplane Shear (Mt) 3 3 3 3

Shear Stress Summary 4 4 3 3

Vessel Stresses (MPa):

LONGITUDINAL PLANE Au Al Bu Bl
 (Stress Intensities Top Top Bottom Bottom
 Outside Inside Outside Inside

Two * Max Shear Stress 40 86 140 111

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stresses Normal to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Outplane Membrane (P) 3 3 3 3
 Outplane Bending (P) 5 -5 5 -5
 Outplane Membrane (Mc) -20 -20 20 20
 Outplane Bending (Mc) -63 63 63 -63
 Outplane Membrane (ML) 0 0 0 0
 Outplane Bending (ML) 0 0 0 0
 Normal Pressure Stress 32 32 32 32

Outplane Stress Summary -41 73 126 -11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stresses parallel to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Inplane Membrane (P) 0 0 0 0
 Inplane Bending (P) 12 -12 12 -12
 Inplane Membrane (Mc) -8 -8 8 8
 Inplane Bending (Mc) -121 121 121 -121
 Inplane Membrane (ML) 0 0 0 0
 Inplane Bending (ML) 0 0 0 0
 Inplane Pressure Stress 65 66 65 66

Inplane Stress Summary -50 168 208 -58

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Shear stress normal to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Outplane Shear (Vc) 0 0 0 0
 Outplane Shear (Vl) 0 0 0 0
 Torsional Shear (Mt) 3 3 3 3

Shear Stress Summary 3 3 4 4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stress Intensities Left Left Right Right
 Outside Inside Outside Inside

Two * Max Shear Stress 51 168 209 58

Nozzle Stresses (MPa):

LONGITUDINAL PLANE Au Al Bu Bl



(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-33	-33	33	33
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	32	31	32
Hoop Stress Summary	0	2	68	69

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-11	-11	11	11
Axial Bending (ML)	-154	154	154	-154
Axial Pressure Stress	15	15	15	15
Axial Stress Summary	-121	131	209	-155

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	7	7	5	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	123	131	209	225

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-20	-20	20	20
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	32	31	32
Hoop Stress Summary	14	15	55	56

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-281	281	281	-281
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0



Axial Pressure Stress 15 15 15 15

Axial Stress Summary -245 261 333 -284

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	5	5	7	7

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	260	261	333	341

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	65	66	65	66	65	66	65	66
Circ. Pl (SUS)	-30	-30	36	36	-8	-8	8	8
Circ. Q (SUS)	-25	25	35	-35	-109	109	133	-133
Long. Pm (SUS)	32	32	32	32	32	32	32	32
Long. Pl (SUS)	-5	-5	5	5	-17	-17	23	23
Long. Q (SUS)	-57	57	81	-81	-58	58	68	-68
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	65.0	66.0	65.0	66.0	65.0	66.0	65.0	66.0
Pm+Pl (SUS)	35.0	36.0	101.0	102.0	57.0	58.0	73.0	74.0
Pm+Pl+Q (Total)	40.4	84.4	136.5	111.2	52.9	167.1	206.1	59.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	66.00	138.00	Passed
Pm+Pl (SUS)	102.00	207.00	Passed
Pm+Pl+Q (TOTAL)	206.11	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	32	31	32	31	32	31	32
Circ. Pl (SUS)	-30	-30	36	36	-17	-17	23	23
Circ. Q (SUS)	0	0	0	0	0	0	0	0



Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-11	-11	11	11	-8	-8	8	8
Long. Q (SUS)	-127	127	181	-181	-254	254	308	-308
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	31.0	32.0	31.0	32.0	31.0	32.0	31.0	32.0
Pm+Pl (SUS)	4.3	4.4	67.0	68.0	14.1	15.1	54.0	55.0
Pm+Pl+Q (Total)	124.8	131.4	207.2	223.2	261.2	261.1	331.2	340.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	138.00	Passed
Pm+Pl (SUS)	68.02	207.00	Passed
Pm+Pl+Q (TOTAL)	340.29	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N6A 2"

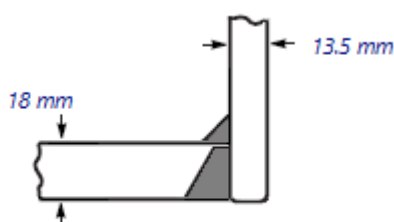
From: 50

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	500.0000	mm
Distance from Bottom/Left Tangent		11700.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-16.43	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: N6A 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 0.2266 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	118.4893	mm
Parallel to Vessel Wall, opening length	d	59.2446	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N6A 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.227, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N6A 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: N6A 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 0.2266 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N6A 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2266	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 16.9877	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 16.9877	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(16.9877, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.2266, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Weld Size Calculations, Description: N6A 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 12.0803 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.8549 mm

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Input, Nozzle Desc: N6B 2"

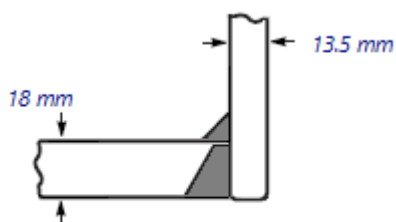
From: 50

Pressure for Reinforcement Calculations	P	1.1031	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		11700.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N6B 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 14.0800$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 0.2281$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N6B 2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2281	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.0800	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.0800	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(17.08, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.2281, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 13.462, tr = 0.228, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.022$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N6B 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 0.4316 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.4316 mm

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Input, Nozzle Desc: N10 1.5"

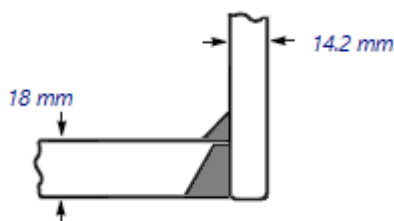
From: 50

Pressure for Reinforcement Calculations	P	1.1015	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	1525.0000	mm
Distance from Bottom/Left Tangent		12000.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-59.60	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: N10 1.5"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.500 in.
 Actual Thickness Used in Calculation 0.560 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 14.0598 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 22.05 \cdot (\exp([1.1 / (138.0 \cdot 1.0)] - 1))$
 $= 0.1767 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	176.8544	mm
Parallel to Vessel Wall, opening length	d	88.4272	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	28.0600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N10 1.5".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govern. thk, tg = 14.224, tr = 0.177, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.016$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N10 1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.2240	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: N10 1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.500 in.
 Actual Thickness Used in Calculation 0.560 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 14.0598 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]



$$= R(\exp([P/(S_n \cdot E)] - 1)) \text{ per Appendix 1-2 (a)(1)}$$

$$= 22.05(\exp([1.1/(138.0 \cdot 1.0)] - 1))$$

$$= 0.1767 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	96.5480	mm
Parallel to Vessel Wall	Rn+tn+t	48.2740	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	28.0600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N10 1.5".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.1767	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.0598	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.0598	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.5200	mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[7.52, max(17.0598, 4.5)]
 = 7.5200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.1767, 7.52)
 = 7.5200 mm

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Weld Size Calculations, Description: N10 1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.2240	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 61.9960 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 348.1204 mm

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Input, Nozzle Desc: N12A 1"

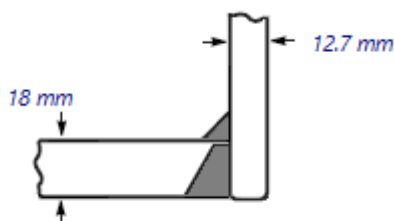
From: 50

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	500.0000	mm
Distance from Bottom/Left Tangent		11400.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-16.43	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: N12A 1"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.000 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([1.1 / (138.0 \cdot 1.0)] - 1))$
 $= 0.1252 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	82.1499	mm
Parallel to Vessel Wall	Rn+tn+t	41.0750	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N12A 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 12.7, tr = 0.125, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.013$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N12A 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.7000	mm
---	------	--------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: N12A 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.000 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 13.9877 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]



= $R(\exp([P/(S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 = $15.7(\exp([1.1/(138.0 \cdot 1.0)] - 1))$
 = 0.1252 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	80.8000 mm
Parallel to Vessel Wall	Rn+tn+t	40.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.2500 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N12A 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.1252 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 16.9877 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 16.9877 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[6.42, max(16.9877, 4.5)]
 = 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.1252, 6.42)
 = 6.4200 mm

Available Nozzle Neck Thickness = 12.7000 mm -> OK

Weld Size Calculations, Description: N12A 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.7000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 7.7832 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 276.5579 mm

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Input, Nozzle Desc: N12B 1"

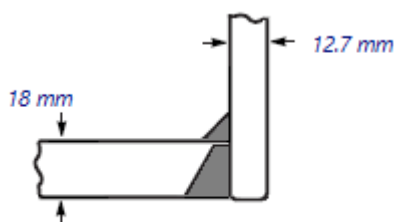
From: 50

Pressure for Reinforcement Calculations	P	1.1031	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		11400.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N12B 1"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.000	in.
Actual Thickness Used in Calculation	0.500	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 14.0800$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 15.7 \cdot (\exp([1.1 / (138.0 \cdot 1.0)] - 1))$
 $= 0.1260$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	80.8000	mm
Parallel to Vessel Wall	Rn+tn+tt	40.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.2500	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N12B 1".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.1260	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 17.0800	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 17.0800	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(17.08, 4.5)]$
 $= 6.4200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.126, 6.42)$
 $= 6.4200$ mm

Available Nozzle Neck Thickness = 12.7000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 12.7, tr = 0.126, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.013$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N12B 1"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.7000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1843 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.1844 mm

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Input, Nozzle Desc: N3 8"

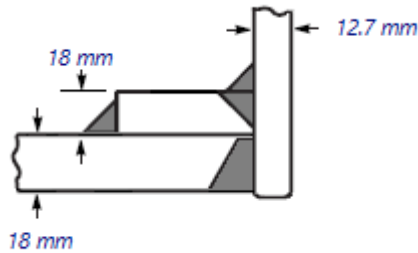
From: 60

Pressure for Reinforcement Calculations	P	1.1031	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12360.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	359.0750	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N3 8"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	7.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 14.0800 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 99.84 / (118 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 0.9386 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500 mm
Parallel to Vessel Wall, opening length	d	199.6750 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

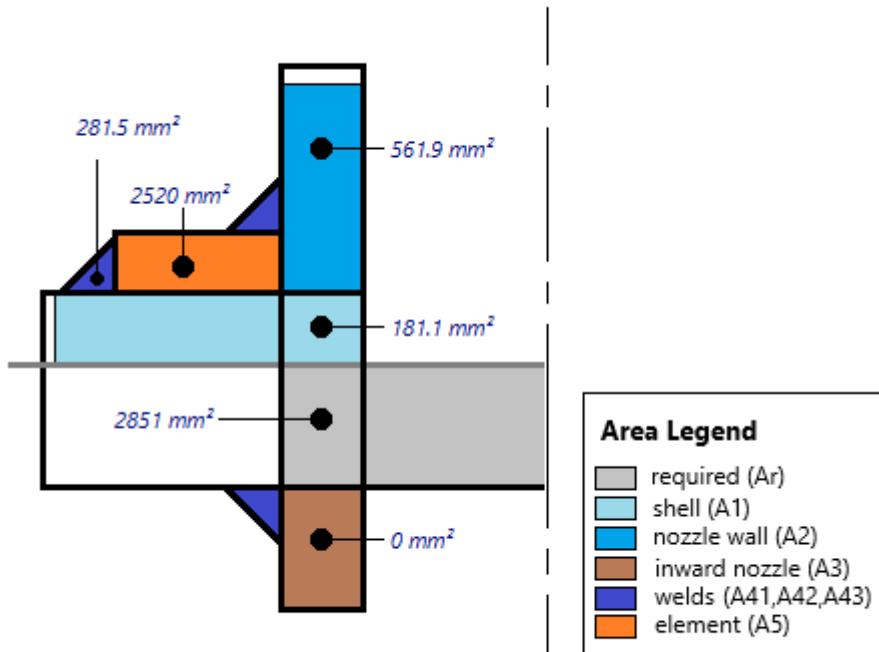


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2851.004	NA	NA
Area in Shell	A1	181.117	NA	NA
Area in Nozzle Wall	A2	561.886	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	281.509	NA	NA
Area in Element	A5	2520.000	NA	NA
TOTAL AREA AVAILABLE	Atot	3544.512	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 320.5468 18.0000 mm
 Based on given Pad Diameter: 359.0750 13.0464 mm
 Based on Shell or Nozzle Thickness: 362.8933 12.7000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (199.675 \cdot 14.08 \cdot 1.0 + 2 \cdot 9.7 \cdot 14.08 \cdot 1.0 \cdot (1 - 0.86))$
 $= 2851.004 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 199.675(1.0 \cdot 15.0 - 1.0 \cdot 14.08) - 2 \cdot 9.7$
 $(1.0 \cdot 15.0 - 1.0 \cdot 14.08) \cdot (1 - 0.855)$
 $= 181.117 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (9.7 - 0.94) \cdot 0.8551$

= 561.886 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 14.0^2 \cdot 1.0$
 = 281.509 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(359.075 - 219.075) \cdot 18.0 \cdot 1.0$
 = 2520.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9386 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 17.0800 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 17.0800 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[10.16, max(17.08, 4.5)]
 = 10.1600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.9386, 10.16)
 = 10.1600 mm

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	29.6,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	265.4 MPa	Passed
Occasional	:	5.4,	Allowable	:	156.9 MPa	Passed
Shear	:	16.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.113, tr = 0.939, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.116$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.939$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 11.113, tr = 0.939, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E^* / (tg - c) = 0.116$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -29 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $1.10/5.10 = 0.216$

Weld Size Calculations, Description: N3 8"

Intermediate Calc. for nozzle/shell Welds Tmin 9.7000 mm
 Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	7.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2851.0044 - 181.1171 + 2 \cdot 9.7 \cdot 0.8551 \cdot (1.0 \cdot 15.0 - 14.08)) \cdot 138)$
 $= 370512.38 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (561.8856 + 2520.0 + 281.5088 - 0.0 \cdot 0.86) \cdot 138$
 $= 464100.59 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (561.8856 + 0.0 + 85.5088 + (248.8306)) \cdot 138$
 $= 123666.30 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (561.8856 + 0.0 + 281.5088 + 2520.0 + (248.8306)) \cdot 138$
 $= 498435.69 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 198955. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 359.075 \cdot 14.0 \cdot 0.49 \cdot 138$$

$$= 533904. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 104.6875) \cdot (12.7 - 3.0) \cdot 0.7 \cdot 118$$

$$= 263487. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 219.075 \cdot 18.0 \cdot 0.74 \cdot 138$$

$$= 632498. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 527072. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (533904 + 263487) = 797392 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (198955 + 632498 + 527072 + 0) = 1358525 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (533904 + 527072 + 0) = 1060976 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 797391 N	, must exceed W = 370512 N	or W1 = 464100 N
Path 2-2 = 1358524 N	, must exceed W = 370512 N	or W2 = 123666 N
Path 3-3 = 1060976 N	, must exceed W = 370512 N	or W3 = 498435 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.4316 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 271.4316 mm

Input Echo, WRC297 Item 1, Description: N3 8" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3500.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	193.675	mm



Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	359.075	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1.10	MPa
Radial Load	P	-5400.01	N
Circumferential Shear	Vc	8000.00	N
Longitudinal Shear	Vl	8000.00	N
Circumferential Moment	Mc	0.480E+07	N-mm
Longitudinal Moment	Ml	0.640E+07	N-mm
Torsional Moment	Mt	0.800E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(1769.5(18.0 - 3.0))}$
 $= 756.707 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3539.000	mm
Nozzle Outside Diameter	(d) =	219.075	mm
Vessel Thickness used	(T) =	33.000	mm
Nozzle Thickness used	(t) =	9.700	mm
T / t	=	3.402	
d / t	=	22.585	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.641$$

Nr/P	=	0.109
Mr/P	=	0.121
M0/P	=	0.131
N0/P	=	0.268
MrD/Mc	=	0.168
NrDL/Mc	=	0.134
M0d/Mc	=	0.225
N0DL/Mc	=	0.096
MrD/Ml	=	0.154
NrDL/Ml	=	0.127
M0D/Ml	=	0.200
N0DL/Ml	=	0.201

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5



Outplane Bending (ML)	-32	32	32	-32
Normal Pressure Stress	58	59	58	59

Outplane Stress Summary	25	83	100	29
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-24	24	24	-24
Inplane Pressure Stress	29	29	29	29
Inplane Stress Summary	4	47	61	4

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	3	3	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	26	83	100	30

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-27	27	27	-27
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	29	29	29	29
Outplane Stress Summary	5	51	63	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-20	20	20	-20
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	58	59	58	59
Inplane Stress Summary	39	73	85	38

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	3	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	39	73	85	38

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	14	12	14
Hoop Stress Summary	8	10	19	21

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	22	-22	22	-22
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-19	-19	19	19
Axial Bending (ML)	-169	169	169	-169
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-159	133	219	-164

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	8	8

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	170	135	219	186

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	14	12	14

Hoop Stress Summary	12	13	16	17
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	22	-22	22	-22
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-140	140	140	-140
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-125	109	185	-140
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10

Shear Stress Summary	8	8	13	13
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	138	110	186	160
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(1760.5(18.0 - 3.0))}$
 $= 755.338 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	3521.000 mm
Nozzle Outside Diameter (d)	=	359.075 mm
Vessel Thickness used (T)	=	15.000 mm
Nozzle Thickness used (t)	=	79.700 mm
T / t	=	0.188
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.562$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.084
Mr/P	=	0.173
M0/P	=	0.050
N0/P	=	0.089
MrD/Mc	=	0.299
NrDL/Mc	=	0.086
M0d/Mc	=	0.092
N0DL/Mc	=	0.066
MrD/M1	=	0.137
NrDL/M1	=	0.099
M0D/M1	=	0.042



NØDL/M1 = 0.107

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	128	129	128	129
Outplane Stress Summary	109	135	166	112

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	17	98	164	-16

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	109	136	167	128

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-32	32	32	-32
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64
Outplane Stress Summary	36	87	110	30

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-106	106	106	-106
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	129	128	129

Inplane Stress Summary	43	207	266	5
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		44	207	266	30

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	58	59	58	59	58	59	58	59
Circ. Pl (SUS)	-4	-4	6	6	-2	-2	2	2
Circ. Q (SUS)	-29	29	35	-35	-17	17	23	-23
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-3	-3	3	3	0	0	2	2
Long. Q (SUS)	-21	21	27	-27	-24	24	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	58.0	59.0	58.0	59.0	58.0	59.0	58.0	59.0
Pm+Pl (SUS)	54.0	55.0	64.0	65.0	56.0	57.0	60.0	61.0
Pm+Pl+Q (Total)	25.4	84.2	99.2	30.4	39.3	74.4	83.4	38.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	59.00	138.00	Passed
Pm+Pl (SUS)	65.00	207.00	Passed
Pm+Pl+Q (TOTAL)	99.22	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	128	129	128	129	128	129	128	129
Circ. Pl (SUS)	-6	-6	10	10	-3	-3	7	7
Circ. Q (SUS)	-13	13	27	-27	-82	82	130	-130
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-5	-5	9	9	-1	-1	5	5
Long. Q (SUS)	-41	41	89	-89	-25	25	39	-39
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	128.0	129.0	128.0	129.0	128.0	129.0	128.0	129.0
Pm+Pl (SUS)	122.0	123.0	138.0	139.0	125.0	126.0	135.0	136.0
Pm+Pl+Q (Total)	109.0	136.1	166.0	128.1	43.7	208.0	265.0	30.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	129.00	138.00	Passed
Pm+Pl (SUS)	139.00	207.00	Passed
Pm+Pl+Q (TOTAL)	265.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	14	12	14	12	14	12	14
Circ. Pl (SUS)	-4	-4	6	6	0	0	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-19	-19	19	19	-14	-14	14	14
Long. Q (SUS)	-147	147	191	-191	-118	118	162	-162
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	12.0	14.0	12.0	14.0	12.0	14.0	12.0	14.0
Pm+Pl (SUS)	21.4	23.3	25.5	25.7	20.4	22.4	20.6	20.8
Pm+Pl+Q (Total)	169.7	135.2	216.3	186.7	138.9	110.7	182.9	159.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+Pl (SUS)	25.70	177.00	Passed
Pm+Pl+Q (TOTAL)	216.32	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.





Input, Nozzle Desc: N8 3"

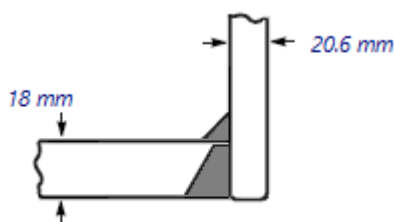
From: 60

Pressure for Reinforcement Calculations	P	1.1031	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3500.00	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		12770.00	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N8 3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.810	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 1753.0 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 14.0800 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 0.3301 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 164.4000 mm
 Parallel to Vessel Wall, opening length d 82.2000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

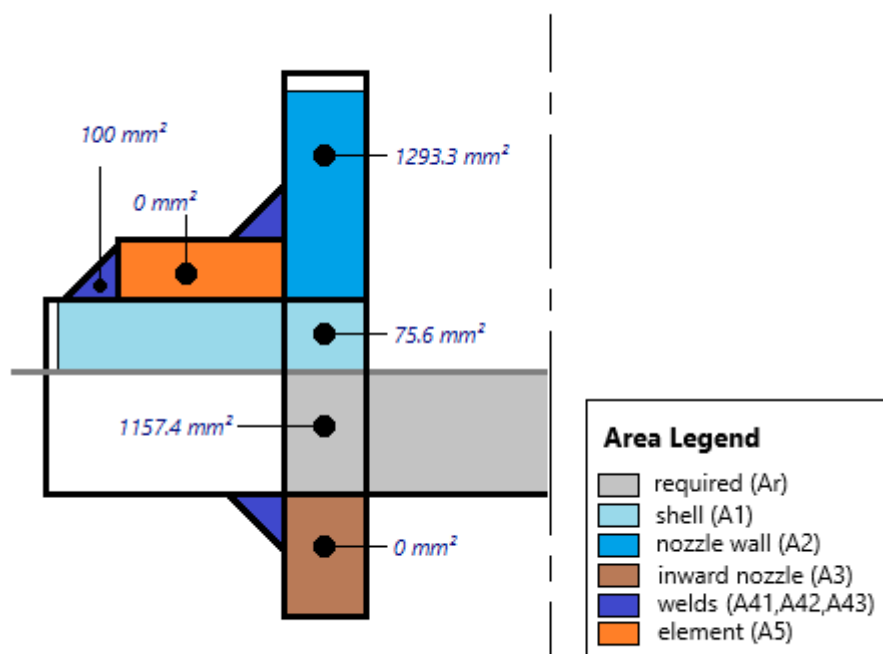


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1157.375	NA	NA
Area in Shell	A1	75.625	NA	NA
Area in Nozzle Wall	A2	1293.292	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	100.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	1468.917	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (82.2 \cdot 14.08 \cdot 1.0 + 2 \cdot 17.574 \cdot 14.08 \cdot 1.0 \cdot (1 - 1.0))$
 $= 1157.375 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 82.2(1.0 \cdot 15.0 - 1.0 \cdot 14.08) - 2 \cdot 17.574$
 $(1.0 \cdot 15.0 - 1.0 \cdot 14.08) \cdot (1 - 1.0)$
 $= 75.625 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 37.5)(17.57 - 0.33)1.0$
 $= 1293.292 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.3301 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 17.0800 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 17.0800 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 8.7300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.73, \max(17.08, 4.5)]$
 $= 8.7300 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3301, 8.73)$
 $= 8.7300 \text{ mm}$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 9.7,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 335.3 MPa	Passed
Occasional	: 1.1,	Allowable	: 183.5 MPa	Passed
Shear	: 7.8,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 14.08, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr * E / (tg - c) = 0.939$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: N8 3"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (1157.375 - 75.625 + 2 \cdot 17.574 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 14.08)) \cdot 138)$$

$$= 153728.09 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (1293.292 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 192254.45 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (1293.292 + 0.0 + 100.0 + (527.22)) \cdot 138$$

$$= 265003.31 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (1293.292 + 0.0 + 100.0 + 0.0 + (527.22)) \cdot 138$$

$$= 265003.31 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 117.348 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 124631. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 49.887) \cdot (20.574 - 3.0) \cdot 0.7 \cdot 138$$

$$= 266041. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 117.348 \cdot (18.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 282332. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (124631 + 266041) = 390672 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (124631 + 0 + 282332 + 0) = 406963 \text{ N}$$

Failure Path [Path 3-3]:



$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (124631 + 282332 + 0) = 406963 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 390671 N , must exceed W = 153728 N or W1 = 192254 N
 Path 2-2 = 406963 N , must exceed W = 153728 N or W2 = 265003 N
 Path 3-3 = 406963 N , must exceed W = 153728 N or W3 = 265003 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.9839 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.9839 mm

Input Echo, WRC297 Item 1, Description: N8 3" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3500.000	mm
Shell Thickness	18.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	76.200	mm
Nozzle Thickness	20.5740	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1.10	MPa
Radial Load	P	-2000.00	N
Circumferential Shear	Vc	3000.00	N
Longitudinal Shear	Vl	3000.00	N
Circumferential Moment	Mc	600000.00	N-mm
Longitudinal Moment	Ml	900000.06	N-mm
Torsional Moment	Mt	0.110E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.704) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 117.348 + 2 \cdot 1.65 \cdot \sqrt{(1760.5(18.0 - 3.0))}$
 $= 653.611 \text{ mm}$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 3521.000 mm
 Nozzle Outside Diameter (d) = 117.348 mm
 Vessel Thickness used (T) = 15.000 mm
 Nozzle Thickness used (t) = 17.574 mm
 T / t = 0.854
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.511$$

Nr/P = 0.164
 Mr/P = 0.290
 M0/P = 0.097
 N0/P = 0.172
 MrD/Mc = 0.297
 NrDL/Mc = 0.125
 M0d/Mc = 0.112
 N0DL/Mc = -0.079
 MrD/Ml = 0.262
 NrDL/Ml = 0.123
 M0D/Ml = 0.097
 N0DL/Ml = 0.006

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	128	129	128	129
Outplane Stress Summary	114	145	155	106

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-53	53	53	-53
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	23	99	138	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	115	145	155	106



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-15	15	15	-15
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64
Outplane Stress Summary	57	77	84	43

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-40	40	40	-40
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	129	128	129
Inplane Stress Summary	101	153	188	77

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	102	153	188	78

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3
Hoop Stress Summary	3	5	4	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0



Axial Bending (P)	8	-8	8	-8
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-29	29	29	-29
Axial Pressure Stress	1	1	1	1

Axial Stress Summary -27 15 47 -28

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary 3 3 1 1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	32	17	47	34

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary 5 7 2 3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	8	-8	8	-8
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-23	23	23	-23
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary -18 11 37 -24

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary 1 1 3 3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	24	12	38	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	128	129	128	129	128	129	128	129
Circ. Pl (SUS)	1	1	1	1	-1	-1	3	3
Circ. Q (SUS)	-14	14	24	-24	-25	25	55	-55
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-3	-3	5	5	2	2	0	0
Long. Q (SUS)	-38	38	68	-68	-10	10	20	-20
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	128.0	129.0	128.0	129.0	128.0	129.0	128.0	129.0
Pm+Pl (SUS)	129.0	130.0	129.0	130.0	127.0	128.0	131.0	132.0
Pm+Pl+Q (Total)	115.2	144.4	153.2	106.0	102.1	153.1	186.2	77.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	129.00	138.00	Passed
Pm+Pl (SUS)	132.01	207.00	Passed
Pm+Pl+Q (TOTAL)	186.16	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	1	1	1	1	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-7	-7	7	7	-4	-4	4	4
Long. Q (SUS)	-21	21	37	-37	-15	15	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	9.0	10.0	8.0	8.0	7.0	8.0	5.0	5.0
Pm+Pl+Q (Total)	30.3	15.4	45.1	33.2	22.4	12.5	36.1	29.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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PV Elite 28 Licensee: SL User (Modify in Config->Save Options)
FileName : 405-V-003_7% Page 176 of 186
Nozzle Calcs.: N8 3" Nozl: 46 2:37pm Jul 20,2026



Pm (SUS)	3.00	138.00	Passed
Pm+Pl (SUS)	10.00	207.00	Passed
Pm+Pl+Q (TOTAL)	45.10	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: M1 24"

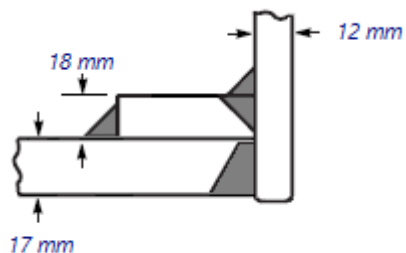
From: 70

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3502.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	17.0000	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1033.6000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		200.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M1 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.1 \cdot 3508.0 \cdot 0.8985 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 1.1)$
 $= 12.5256 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 307.8 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 2.4560 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1231.2001	mm
Parallel to Vessel Wall, opening length	d	615.6000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		35.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

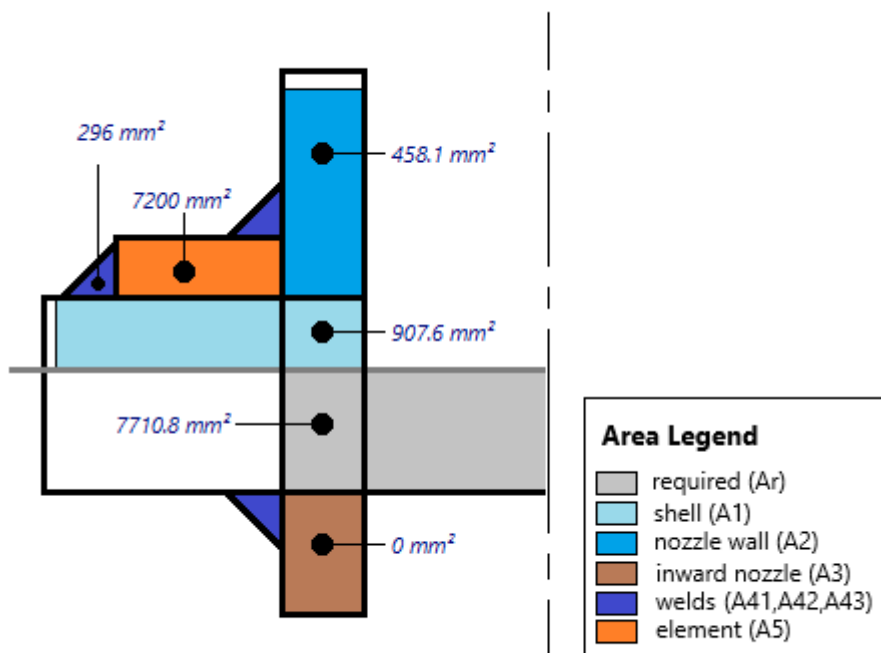


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7710.761	NA	NA
Area in Shell	A1	907.640	NA	NA
Area in Nozzle Wall	A2	458.081	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	296.000	NA	NA
Area in Element	A5	7200.000	NA	NA
TOTAL AREA AVAILABLE	Atot	8861.721	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 969.6578 18.0000 mm
 Based on given Pad Diameter: 1033.6000 15.1226 mm
 Based on Shell or Nozzle Thickness: 1137.6868 12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (615.6 \cdot 12.5256 \cdot 1.0 + 2 \cdot 9.0 \cdot 12.5256 \cdot 1.0 \cdot (1 - 1.0))$
 $= 7710.761 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 615.6(1.0 \cdot 14.0 - 1.0 \cdot 12.526) - 2 \cdot 9.0$
 $(1.0 \cdot 14.0 - 1.0 \cdot 12.526) \cdot (1 - 1.0)$
 $= 907.640 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 35.0) \cdot (9.0 - 2.46) \cdot 1.0$



= 458.081 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 14.0^2 \cdot 1.0$
 = 296.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1033.6 - 633.6) \cdot 18.0 \cdot 1.0$
 = 7200.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.4560$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = t_a
 = 5.4560 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.456$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.273$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 17.0$, $t_r = 12.526$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.895$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 2.456$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.273$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-71 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :



Design Pressure/Ambient Rating = 1.10/1.96 = 0.559

Weld Size Calculations, Description: M1 24"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	8.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (7710.7612 - 907.6395 + 2 \cdot 9.0 \cdot 1.0 \cdot (1.0 \cdot 14.0 - 12.5256)) \cdot 138)$
 $= 942396.00 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (458.0812 + 7200.0 + 296.0 - 0.0 \cdot 1.0) \cdot 138$
 $= 1097550.12 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (458.0812 + 0.0 + 100.0 + (252.0)) \cdot 138$
 $= 111779.67 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (458.0812 + 0.0 + 296.0 + 7200.0 + (252.0)) \cdot 138$
 $= 1132322.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 633.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 672935. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1033.6 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 1536848. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 312.3) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 852913. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 633.6 \cdot 18.0 \cdot 0.74 \cdot 138$
 $= 1829285. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 633.6 \cdot (17.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 1422778. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (1536848 + 852913) = 2389760 \text{ N}$



Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (672935 + 1829285 + 1422778 + 0) = 3924998 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (1536848 + 1422778 + 0) = 2959625 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2389760 N	, must exceed W = 942396 N	or W1 = 1097550 N
Path 2-2 = 3924998 N	, must exceed W = 942396 N	or W2 = 111779 N
Path 3-3 = 2959625 N	, must exceed W = 942396 N	or W3 = 1132322 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 15.8928 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 282.8928 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 1.931 %

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Input, Nozzle Desc: N7 3"

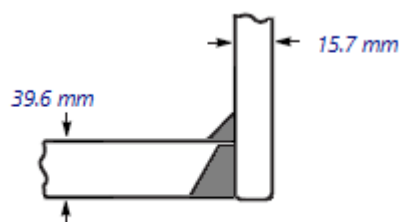
From: 80

Pressure for Reinforcement Calculations	P	1.0959	MPa
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Outside Diameter of Bolted Blind Flange	D	381.00	mm
Head Finished (Minimum) Thickness	t	39.6240	mm
Head Internal Corrosion Allowance	c	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	15.7480	mm
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	39.6240	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N7 3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.620	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.1 \cdot 41.1 / (138 \cdot 1.0 - 0.6 \cdot 1.1)$
 $= 0.3279 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 180.9440 mm
 Parallel to Vessel Wall Rn+tn+t 90.4720 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 31.8700 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 138.0/138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

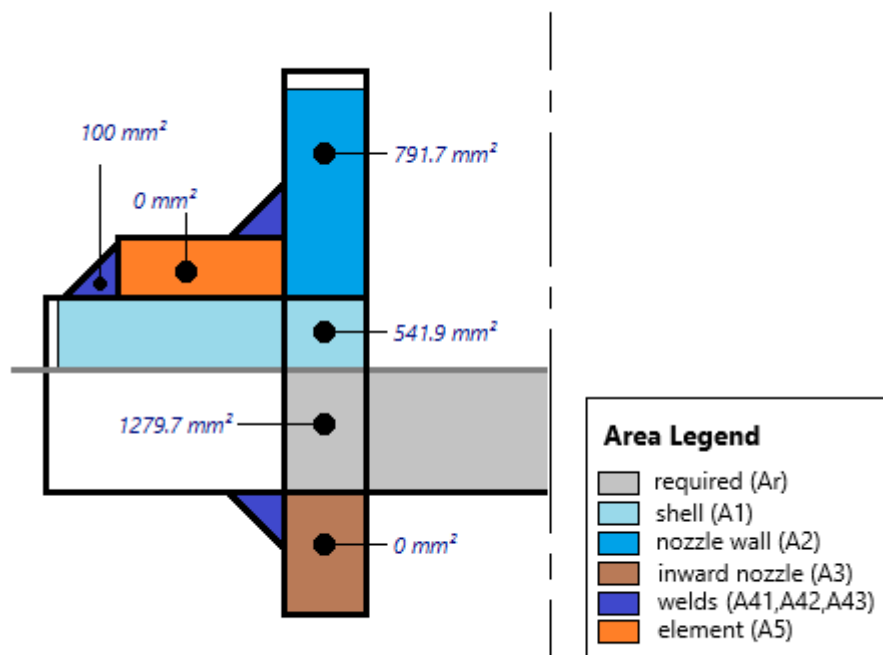


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1279.710	1156.570	NA
Area in Shell	A1	541.857	837.704	NA
Area in Nozzle Wall	A2	791.654	812.557	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	1433.511	1750.262	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5 \cdot d \cdot t + t \cdot \ln(1 - fr_1)$ per UG-39(b)(1)
 $= 0.5 \cdot 82.2 \cdot 31.1365 + 31.1365 \cdot 12.748(1 - 1.0)$
 $= 1279.710 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E_1 t - F \cdot tr) - 2 \cdot \ln(E_1 t - F \cdot tr) \cdot (1 - fr_1)$
 $= 98.744(1.0 \cdot 36.624 - 1.0 \cdot 31.137) - 2 \cdot 12.748$
 $(1.0 \cdot 36.624 - 1.0 \cdot 31.1365) \cdot (1 - 1.0)$
 $= 541.857 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot \ln p)(\ln - \ln) fr_2$
 $= (2 \cdot 31.87)(12.75 - 0.33) 1.0$
 $= 791.654 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$ta = 3.3279 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 34.1365 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 34.1365 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.258, \max(34.1365, 4.5)]$
 $= 8.2578 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.3279, 8.2578)$
 $= 8.2578 \text{ mm}$

Available Nozzle Neck Thickness = 15.7480 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 15.748$, $tr = 0.328$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.026$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N7 3"

Intermediate Calc. for nozzle/shell Welds Tmin 12.7480 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1279.7103 - 541.8569 + 2 \cdot 12.748 \cdot 1.0 \cdot (1.0 \cdot 36.624 - 31.1365)) \cdot 138)$$

$$= 121118.74 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (791.6544 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 138$$

$$= 123035.62 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (791.6544 + 0.0 + 100.0 + (933.7654)) \cdot 138$$

$$= 251881.97 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (791.6544 + 0.0 + 100.0 + 0.0 + (933.7654)) \cdot 138$$

$$= 251881.97 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 107.696 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 114380. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 47.474) \cdot (15.748 - 3.0) \cdot 0.7 \cdot 138$$

$$= 183649. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 107.696 \cdot (39.624 - 3.0) \cdot 0.74 \cdot 138$$

$$= 632644. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (114380 + 183649) = 298029 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (114380 + 0 + 632644 + 0) = 747023 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (114380 + 632644 + 0) = 747023 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 298028 N	, must exceed W = 121118 N	or W1 = 123035 N
Path 2-2 = 747023 N	, must exceed W = 121118 N	or W2 = 251881 N
Path 3-3 = 747023 N	, must exceed W = 121118 N	or W3 = 251881 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

The Cut Length for this Nozzle is, Drop + Ho + H + T : 289.6748 mm

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